



CNS

MIDTERM

MEDICAL HORIZON
MNU'1 - MU'41

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Formatives

1. The main action of buccinator muscle is :

- A. Eye closure.
- B. Mouth closure.
- C. Prevent food accumulation in the mouth vestibule.
- D. Mandible depression

2. The dangerous layer of the scalp is:

- A. Skin layer.
- B. Subcutaneous connective tissue layer.
- C. Epicranial aponeurosis.
- D. Loose connective tissue layer.

3. The mastication muscle which is concerned with mandibular retraction is:

- A. Masseter
- B. Posterior fibers of temporalis.
- C. Anterior fibers of temporalis.
- D. Medial & lateral pterygoids.

4. The strongest ligament of T/M joint is:

- A. Pterygo- mandibular ligament.
- B. Stylo- mandibular ligament.
- C. Spheno- mandibular ligament.
- D. Lateral ligament of T/M joint.

5. It leaves the skull through the stylo- mastoid foramen:

- A. Maxillary nerve.
- B. Mandibular nerve.
- C. Ophthalmic nerve.
- D. Facial nerve.

6. The common facial vein ends into:

- A. Internal jugular vein.
- B. External jugular vein.
- C. Anterior jugular vein.
- D. Posterior jugular vein.

7. What is the most internal layer of the scalp?

- A. Epicranial aponeurosis
- B. Periosteum
- C. Dense connective tissue
- D. Dura mater

8. Contraction of left lateral pterygoid will result in:

- A. Deviation of mandible upwards
- B. Deviation of mandible downwards
- C. Deviation of mandible to right side
- D. Deviation of mandible to left side

9. Supranuclear paralysis of the facial motor nucleus will not affect:

- A. Orbicularis oculi
- B. Buccinators
- C. Nasalis
- D. Orbicularis oris

10. Which vein is responsible for draining the majority of the external face?

- A. External jugular vein
- B. Internal jugular vein
- C. Anterior jugular vein
- D. Posterior auricular vein

Answers:

1-C , 2-D , 3-B , 4-D , 5-D , 6-A , 7-B , 8-C , 3-9 , 10-A

Previous Mid

1. The medial pterygoid muscle arises from:

- a) Maxillary tuberosity only
- b) Medial surface of the lateral pterygoid plate only
- c) Two heads: superficial from maxillary tuberosity, deep from medial surface of lateral pterygoid plate
- d) Zygomatic arch

2. The medial pterygoid muscle inserts on which specific part of the mandible?

- A) Lateral surface of the mandibular condyle.
- B) External oblique line.
- C) Medial side of the ramus near its angle.
- D) Mylohyoid line.

3. The deep structures of the face are supplied by:

- a. The superficial temporal arteries.
- b. The facial arteries.
- c. The subclavian arteries.
- d. The maxillary arteries.

4. The medial pterygoid follows the same angle as masseter - arising from the medial pterygoid plate, where does it attach on the mandible?

- a. Ramus and angle.
- b. Condylod process.
- c. Coronoid process.
- d. Mental protuberance.

5. The SVE (special visceral efferent) functional component of the trigeminal nerve is carried by:

- a. Ophthalmic division.
- b. Maxillary division.
- c. Mandibular division.
- d. both "Ophthalmic division" and "Mandibular division".

Answers:

1-c , 2-C , 3-d , 4-a , 5-c

1. In which layer of the scalp would you find the most blood vessels?

- A. Skin
- B. Periosteum
- C. Dense Connective Tissue
- D. Epicranial Aponeurosis

2. How many layers of the scalp are there?

- A. 3
- B. 4
- C. 5
- D. 6

3. What is the action of corrugator supercilii?

- A. Closure of the eyelid
- B. Drawing the eyebrows together
- C. Compression of the nares
- D. Pursing the lips

4. Which of the following muscles opens the nares?

- Procerus
- Transverse part of nasalis
- Alar part of nasalis
- Orbicularis oris

5. Where does the buccinator muscle originate from?

- A. Mandible and maxilla
- B. Nasal septum
- C. Frontal bone
- D. Roof of the orbit

6. Which pharyngeal arch do the muscles of mastication develop from?

- A. First pharyngeal arch
- B. Second pharyngeal arch
- C. Fourth pharyngeal arch
- D. Sixth pharyngeal arch

7. What is the main action of the masseter muscle?

- A. Depression of the mandible
- B. Elevation of the mandible
- C. Retraction of the mandible
- D. Protraction of the mandible

8. Which nerve innervates the muscles of mastication?

- A. Maxillary nerve
- B. Mandibular nerve
- C. Lingual nerve
- D. Glossopharyngeal nerve

9. Where does the temporalis muscle insert?

- A. Coronoid process of the mandible
- B. Corocoid process of the mandible
- C. Neck of the mandible
- D. Ramus of the mandible

10. Which muscle is the medial pterygoid found inferior to?

- A. Masseter
- B. Temporalis

C. Lateral pterygoid

D. Buccinator

11. What is the main action of the lateral pterygoid muscle?

- A. Elevation of the mandible
- B. Depression of the mandible
- C. Retraction of the mandible
- D. Protraction of the mandible

12. Which nerve innervates the orbicularis oculi muscle?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Trigeminal nerve
- D. Lingual nerve

13. Which nerve innervates the orbicularis oris muscle?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Trigeminal nerve
- D. Lingual nerve

14. Which nerve is responsible for the majority of taste sensation?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Trigeminal nerve
- D. Lingual nerve

15. Which nerve is responsible for the majority of the tongue's motor function?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Trochlear nerve
- D. Hypoglossal nerve

16. Which nerve is responsible for general sensation in the anterior 2/3 of the tongue?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Lingual nerve
- D. Hypoglossal nerve

17. Which nerve transmits taste sensation in the posterior 1/3 of the tongue?

- A. Glossopharyngeal nerve
- B. Facial nerve
- C. Trochlear nerve
- D. Hypoglossal nerve

18. Which of the following is not an articulatory surface of the temporomandibular joint?

- A. Mandibular fossa
- B. Articular Tubercle
- C. Mandibular head
- D. Parietal bone

19. Where does the sphenomandibular ligament originate from?

- A. Mandible
- B. Sphenoid spine
- C. Articular tubercle
- D. Mandibular fossa

20. Which of the following muscles is responsible for protrusion of the mandible?

- A. Geniohyoid
- B. Digastric
- C. Temporalis
- D. Lateral pterygoid

21. What mainly contributes to depression of the mandible when there is no resistance?

- A. Digastric
- B. Gravity
- C. Geniohyoid
- D. Mylohyoid

22. At what spinal level does the pharynx end (inferior to the cricoid cartilage)?

- A. C3
- B. C4
- C. C5
- D. C6

23. Which of the following is NOT a part of the pharynx?

- A. Nasopharynx
- B. Oropharynx
- C. Glossopharynx
- D. Laryngopharynx

24. Which of the following is NOT a function of the longitudinal muscles of the pharynx?

- A. Shorten the pharynx
- B. Widen the pharynx
- C. Constrict the lumen of the pharynx
- D. Elevate the larynx

25. Which muscle of the pharynx is the only one not innervated by the vagus nerve (CN X)?

- A. Inferior pharyngeal constrictor
- B. Stylopharyngeus
- C. Palatopharyngeus
- D. Salpingopharyngeus

26. The main action of buccinator muscle is:

- A. Eye closure
- B. Mouth closure
- C. Prevent food accumulation in the mouth vestibule
- D. Mandible depression

27. The dangerous layer of the scalp is:

- A. Skin layer
- B. Subcutaneous connective tissue layer
- C. Epicranial aponeurosis
- D. Loose connective tissue layer

28. They form a potential pathway for spread of infection from the scalp to the intracranial space:

- A. Dural sinuses
- B. Emissary veins
- C. Pterygoid plexus
- D. Facial vein

29. It widens the nares:

- A. Transverse part of nasalis
- B. Alar part of nasalis
- C. Levator anguli oris
- D. Mentalis

30. The 1st pharyngeal arch nerve is:

- A. Facial nerve
- B. Mandibular nerve (V3)
- C. Glossopharyngeal nerve
- D. Vagus nerve

31. It could be used for myringoplasty:

- A. Masseteric fascia
- B. Temporal fascia
- C. Investing fascia of the neck
- D. Prevertebral fascia

32. The only muscle which can retract the protruded mandible:

- A. Masseter
- B. Buccinator
- C. Lateral pterygoid
- D. Temporalis

33. Contraction of right lateral pterygoid would deviate mandible:

- A. Forwards
- B. Backwards
- C. Forwards to the left
- D. Forwards to the right

34. All muscles of the tongue are innervated by CN XII, except:

- A. Palatoglossus
- B. Palatopharyngeus
- C. Tensor palati
- D. Intrinsic tongue muscles

35. Chorda tympani transmits taste sensation from:

- A. Anterior 2/3 of tongue
- B. Anterior 1/3 of tongue
- C. Posterior 2/3 of tongue
- D. Posterior 1/3 of tongue

36. It prevents over opening of mouth:

- A. Pterygomandibular ligament
- B. Stylomandibular ligament
- C. Temporomandibular ligament
- D. Suprahyoid muscles

37. It extends to the lingula of inner surface of mandible:

- A. Pterygomandibular ligament
- B. Stylomandibular ligament
- C. Temporomandibular ligament
- D. Sphenomandibular ligament

38. Its lower head originated from the maxillary tuberosity:

- A. Masseter
- B. Temporalis
- C. Lateral pterygoid
- D. Medial pterygoid

39. All pharyngeal muscles are innervated by vagus nerve except:

- A. Stylopharyngeus
- B. Palatopharyngeus
- C. Salpingopharyngeus
- D. Superior constrictor

Answers :

1-C , 2-C , 3-B , 4-C , 5-A , 6-A , 7-B , 8-B , 9-A ,
10-C , 11-D , 12-B , 13-B , 14-B , 15-D , 16-C ,
17-A , 18-D , 19-B , 20-D , 21-B , 22-D , 23-C ,
24-C , 25-B , 26-C , 27-D , 28-B , 29-B , 30-B ,
31-B , 32-D , 33-C , 34-A , 35-A , 36-C , 37-D ,
38-D , 39-A

Formatives

1. Which of the following cranial nerves can be found in the anterior triangle?

- a) Oculomotor (CN III)
- b) Trochlear (CN IV)
- c) Vestibulocochlear (CN VIII)
- d) Vagus (CN X)

2. Which of the following infrahyoid muscles are found in deep plane?

- a) Sternothyroid and thyrohyoid
- b) Sternohyoid and omohyoid
- c) Sternohyoid and sternothyroid
- d) Sternohyoid and thyrohyoid

3. The nerve supply of the mylohyoid muscle is:

- a) Facial nerve
- b) Mandibular nerve
- c) Maxillary nerve
- d) Ophthalmic nerve

4. Subclavian vessels are contents of:

- a) Submental triangle
- b) Digastric triangle
- c) Posterior triangle
- d) Muscular triangle

5. Two months infant holds his head tilted to one side and has difficulty turning the head to the opposite side. He was diagnosed as congenital torticollis, which muscle is affected?

- a) Sternohyoid
- b) Omohyoid
- c) Sternothyroid
- d) Sternomastoid

6. Regarding the triangles of the neck:

- a) Posterior triangle is bounded by trapezius, sternomastoid and mandible
- b) Anterior triangle is divided into two triangles
- c) Posterior belly of digastric separates the carotid and submandibular
- d) Accessory nerve divides the posterior triangle into lower small and upper

7. Carotid triangle contain:

- a) Glossopharyngeal nerve
- b) Hypoglossal nerve
- c) Facial nerve
- d) Mandibular nerve

Answers:

1-d , 2-a , 3-b , 4-c , 5-d , 6-c , 7-b

1. Which of the following is NOT a border of the anterior triangle?

- a. Midline of neck
- b. Mandible

- c. Sternocleidomastoid
- d. Sternohyoid

2. Which of the following cranial nerves can be found in the anterior triangle?

- a. Olfactory (CNI)
- b. Oculomotor (CN III)
- c. Trochlear (CN IV)
- d. Vagus (CN X)

3. Which important vascular structure bifurcates within the carotid triangle?

- a. Internal jugular vein
- b. Common carotid artery
- c. Subclavian vein
- d. Facial artery

4. Which triangle does the facial artery and vein pass through?

- a. Submental Triangle
- b. Submandibular Triangle
- c. Muscular Triangle
- d. Carotid Triangle

5. What is the superior border of the muscular triangle?

- a. Superior belly of omohyoid
- b. Inferior portion of sternocleidomastoid
- c. Hyoid bone
- d. Imaginary midline of the neck

6. Which muscle forms the lateral border of the anterior triangle?

- a. Trapezius
- b. Levator scapulae
- c. Sternocleidomastoid
- d. Platysma

7. What structure forms the anterior border of the posterior triangle?

- a. Inferior aspect of the mandible
- b. Posterior aspect of the SCM
- c. Anterior aspect of the trapezius
- d. Middle third of the clavicle

8. Which muscle divides the posterior triangle into two?

- a. Anterior scalene
- b. Superior belly of omohyoid
- c. Posterior scalene
- d. Inferior belly of omohyoid

9. Which of these blood vessels is NOT found in the posterior triangle?

- a. Internal jugular vein
- b. External jugular vein
- c. Transverse cervical artery
- d. Subclavian artery

10. Which part of the brachial plexus crosses the floor of the posterior triangle?

- a. Branches
- b. Trunks
- c. Roots
- d. Divisions

11. Which cranial nerve crosses the posterior triangle within the investing layer of fascia?

- a. Facial (CN VII)
- b. Glossopharyngeal (CN IX)
- c. Vagus (CN X)
- d. Accessory (CN XI)

12. Which layer of fascia covers the posterior triangle?

- a. Prevertebral
- b. Visceral
- c. Investing
- d. Superficial cervical

13. Name the muscle that makes up the posterior border of the posterior triangle:

- a. Sternocleidomastoid
- b. Trapezius
- c. Levator scapulae
- d. Middle scalene

14. Which muscle forms part of the floor in the posterior triangle?

- a. Trapezius
- b. Middle scalene
- c. Latissimus dorsi
- d. Serratus posterior inferior

15. The following nerves are present in the anterior triangle, except:

- a. Abducent nerve
- b. Facial nerve
- c. Hypoglossal nerve
- d. Accessory nerve

16. Thyroid gland is a content of:

- a. Submental triangle
- b. Digastric (submandibular) triangle
- c. Carotid triangle
- d. Muscular triangle

17. The fascia of the floor of the posterior triangle is formed by:

- a. Investing neck fascia
- b. Prevertebral fascia
- c. Pretracheal fascia
- d. Sibson's fascia

18. Trunks of the brachial plexus are content of:

- a. Submental triangle
- b. Digastric (submandibular) triangle
- c. Carotid triangle
- d. Posterior triangle

19. Which of the following cranial nerves can be found in the anterior triangle?

- a. Oculomotor (CN III)
- b. Trochlear (CN IV)
- c. Vestibulocochlear
- d. Vagus

Answers:

1-D , 2-D , 3-B , 4-d , 5-C , 6-C , 7-B , 8-D , 9-A ,
10-B , 11-D , 12-C , 13-B , 14-B , 15-A , 16-D ,
17-B , 18-D , 19-D

Formatives

1. The nerve supply to the posterior belly of digastrics is :

- A. Facial nerve.
- B. Mandibular nerve
- C. Maxillary nerve
- D. Ophthalmic nerve.

2. Ptosis and lateral squint are features of injury :

- A. Abducent nerve.
- B. Optic nerve.
- C. Oculo-motor nerve.
- D. Trochlear nerve.

3. Chorda tympani nerve joins nerve in the infra-temporal fossa:

- A. Lingual nerve
- B. Inferior alveolar nerve.
- C. Buccal nerve.
- D. Nerve to masseter

4. Auriculo-temporal nerve is a branch of:

- A. Maxillary nerve
- B. Mandibular nerve.
- C. Ophthalmic nerve.
- D. Facial nerve.

5. Infraorbital nerve is a continuation of:

- A. Maxillary nerve.
- B. Mandibular nerve.
- C. Ophthalmic nerve.
- D. Facial nerve.

6. It leaves the skull through the stylo- mastoid foramen:

- A. Maxillary nerve.
- B. Mandibular nerve.
- C. Ophthalmic nerve.
- D. Facial nerve.

7. Otic ganglion is the site of relay of the parasympathetic fibers that supply:

- A. Lacrimal gland.
- B. Parotid gland.
- C. Submandibular gland.
- D. Sublingual gland.

8. The extra-ocular muscle which is supplied by the trochlear nerve:

- A. Superior rectus.
- B. Inferior rectus.
- C. Inferior oblique.
- D. Superior oblique

9. The parasympathetic supply of the lacrimal gland is derived from:

- A. Ciliary ganglion.
- B. Pterygo-palatine (Spheno-palatine) ganglion.
- C. Otic ganglion.
- D. Submandibular ganglion.

Answers

1-A , 2-C , 3-A, 4-B , 5-A , 6-D , 7-B , 8-D , 9-B

Mid (39)

1. Oculomotor nerve originate from:

- a. Dorsal nucleus
- B. Edinger westphal nucleus
- C. Solatiry nucleus
- D. Spinal nucleus

2. From Nasociliary branche of ophthalmic nerve:

- A. Frontal nerve
- B. Lacrimal nerve
- C. Anterior+posterior ethmoidal nerves
- D. All of above

Answers:

1-B , 2-C

1. The oculomotor nerve exits the cranium through which structure?

- A. Jugular foramen
- B. Superior orbital fissure
- C. Internal acoustic meatus
- D. Foramen ovale

2. Which of the following would NOT test the sensory function of the ophthalmic nerve (CNV1)?

- A. Touching the cornea with a cotton swab
- B. Touching the jaw with a cotton swab
- C. Touching the forehead with a pin
- D. Touching the bridge of the nose with a pin

3. The facial nerve travels through the internal acoustic meatus. Which bone of the skull is this located in?

- A. Sphenoid
- B. Parietal
- C. Temporal
- D. Occipital

4. SVM component of facial nerve is concerned with:

- A. Supplying facial muscles.
- B. Supplying lacrimal gland.
- C. Taste from anterior 2/3 of the tongue.
- D. Sensation from ear.

Answers:

1-B , 2-B , 3-C , 4-A

Formatives

1. The skin area over the angle of the mandible is supplied by

- A. Supra-orbital nerve.
- B. Great auricular nerve
- C. Mandibular nerve.
- D. Maxillary nerve

2. Which of the following cranial nerves can be found in the anterior triangle?

- A. Oculomotor (CN III)
- B. Trochlear (CN IV)
- C. Vestibulocochlear
- D. Vagus

3. Types of fibers present in hypoglossal nerve:

- A. General somatic motor (GSM)
- B. General visceral motor (GVM)
- C. Special visceral motor (SVM)
- D. Special visceral sensory (SVS)

Answers:

1-B , 2-D , 3-A

Mid (37)

1. Descendes cervicalis arises from:

- a. C1+C2
- B. C2+C3
- C. C3+C4
- D. C4+C5

1. Types of fibers present in hypoglossal nerve:

- A. General somatic motor (GSM)
- B. General visceral motor (GVM)
- C. Special visceral motor (SVM)
- D. Special visceral sensory (SVS)

2. What are the functions of the vagus nerve?

- A. Motor
- B. Sensory
- C. Parasympathetic
- D. All of the above

3. What extrinsic tongue muscle is not innervated by the hypoglossal nerve?

- A. Genioglossus
- B. Hypoglossus
- C. Palatoglossus
- D. Styloglossus

4. Which of the following cranial nerves does NOT travel through the jugular foramen?

- A. Glossopharyngeal nerve
- B. Vagus nerve
- C. Accessory nerve
- D. Hypoglossal nerve

5. Where in the brainstem does the glossopharyngeal nerve arise?

- A. Pons
- B. Medulla oblongata
- C. Midbrain
- D. Cerebellum

6. Of the following, which cranial nerve arises from the Pons?

- A. Trochlear
- B. Trigeminal
- C. Vagus
- D. Hypoglossal

7. Which of these structures are innervated by the accessory nerve?

- A. Anterior belly of digastric
- B. Latissimus dorsi
- C. Cricopharyngeus
- D. Trapezius

8. The cervical plexus is made up of the anterior rami of which spinal nerves?

- A. C1-C3
- B. C2-C4
- C. C1-C4
- D. C1-C6

9. The hypoglossal nerve supplies the following muscles, except:

- a. Intrinsic muscles of the tongue
- b. Hyoglossus
- c. Genioglossus
- d. Palatoglossus

10. A malignant tumour is damaging the patient's glossopharyngeal nerve. They will experience:

- a. Loss of taste over the anterior two-thirds of the tongue.
- b. Loss of somesthetic sensation over the anterior two thirds of the tongue.
- c. Loss of taste and somesthetic sensation over the posterior third of the tongue.
- d. Paralysis of the muscles of the tongue.

Answers:

Mid : 1-B

1-A , 2-D , 3-C , 4-D , 5-B , 6-B , 7-D , 8-C , 9-D , 10-C

Formatives

1. It arises from the medial aspect of the external carotid artery:

- A. Facial artery
- B. Lingual artery
- C. Superior thyroid artery
- D. Ascending pharyngeal artery

2. It's a branch of the 3rd part of maxillary artery:

- A. Middle meningeal artery
- B. Inferior alveolar artery
- C. Infra-orbital artery
- D. Masseteric artery

3. Thyro-cervical trunk arises from:

- A. 1st part of subclavian artery
- B. 2nd part of subclavian artery
- C. 3rd part of subclavian artery
- D. 4th part of subclavian artery

4. The common facial vein ends into:

- A. Internal jugular vein
- B. External jugular vein
- C. Anterior jugular vein
- D. Posterior jugular vein

5. Which vertebral arteries arise from which artery?

- A. External carotid artery
- B. Thyrocervical trunk
- C. Subclavian artery
- D. Common carotid artery

6. It arises from the medial aspect of the external carotid artery:

- A. Facial artery
- B. Lingual artery
- C. Superior thyroid artery
- D. Ascending pharyngeal artery

Answers: 1-D , 2-C , 3-A , 4-A , 5-C , 6-D

Mid

1. All of the following are anterior branches of the external carotid artery, EXCEPT:

- A. Superior thyroid artery
- B. Lingual artery
- C. Facial artery
- D. Occipital artery

2. Which of the following is a branch of the 2nd part of the subclavian artery?

- A. Vertebral artery
- B. Thyro-cervical trunk
- C. Internal thoracic artery
- D. Costo-cervical trunk

3. The deep structures of the face are supplied by:

- A. Superficial temporal arteries
- B. Facial arteries
- C. Subclavian arteries
- D. Maxillary arteries

4. External jugular vein terminates by draining into:

- A. Posterior auricular vein
- B. Retromandibular vein (posterior branch)
- C. Retromandibular vein (anterior branch)
- D. Subclavian vein

Answer: 1.D , 2.D , 3.D, 4.D

1. From which artery does the left common carotid artery arise?

- A. Left subclavian artery
- B. Brachiocephalic trunk
- C. Arch of the aorta
- D. Left external carotid artery

2. At which vertebral level do the common carotids bifurcate?

- A. C2
- B. C3
- C. C4
- D. C5

3. Which of the following is a branch of the external carotid artery?

- A. Superficial temporal
- B. Supraorbital
- C. Supratrochlear
- D. Inferior thyroid

4. Which artery gives off the vertebral artery?

- A. Thyrocervical trunk
- B. Subclavian artery
- C. Middle meningeal artery
- D. External carotid artery

5. Which vein is responsible for draining the majority of the external face?

- A. External jugular vein
- B. Internal jugular vein
- C. Anterior jugular vein
- D. Posterior auricular vein

6. Into which vein do the anterior jugular veins empty?

- A. Internal jugular vein
- B. External jugular vein
- C. Subclavian vein
- D. Inferior vena cava

7. The internal jugular vein is a continuation of which sinus?

- A. Cavernous sinus
- B. Inferior petrosal sinus
- C. Sigmoid sinus
- D. Transverse sinus

8. The internal jugular vein combines with which other vein to form the brachiocephalic vein?

- A. External jugular vein
- B. Subclavian vein
- C. Lingual vein
- D. Posterior auricular vein

9. Which lymph node collects lymph from the central lower lip?

- A. Submental
- B. Submandibular
- C. Occipital
- D. Parotid

10. Which of the following is not a deep cervical lymph node?

- A. Prelaryngeal
- B. Pre-auricular
- C. Jugulodigastric
- D. Jugulo-omohyoid

11. They are all branches of thyrocervical trunk except:

- A. Suprascapular artery
- B. Transverse cervical artery
- C. Deep cervical artery
- D. Inferior thyroid artery

12. It is formed by union of maxillary & superficial temporal veins:

- A. Facial vein
- B. Retromandibular vein
- C. EJV
- D. Subclavian vein

13. They drain the anterior part of the floor of mouth & the apex of tongue:

- A. Mastoid LNs
- B. Parotid LNs
- C. Submandibular LNs
- D. Submental LNs

14. Which vertebral arteries arise from which artery?

- A. External carotid
- B. Thyrocervical
- C. Subclavian
- D. Common carotid

15. Which vein is responsible for draining the majority of the external face?

- A. External jugular vein
- B. Internal jugular vein
- C. Anterior jugular vein
- D. Posterior auricular vein

Answers:

1-C , 2-C , 3-A , 4-B , 5-B , 6-B , 7-C , 8-B , 9-A , 10-B , 11-C , 12-B , 13-D , 14-C , 15-B

Formatives

1. Muscles of mastication are derivative of:

- A. 1st pharyngeal arch
- B. 2nd pharyngeal arch
- C. 3rd pharyngeal arch
- D. 4th pharyngeal arch

2. Facial nerve is the nerve component of:

- A. 1st pharyngeal arch
- B. 2nd pharyngeal arch
- C. 3rd pharyngeal arch
- D. 4th pharyngeal arch

3. Unilateral type of cleft lip is a result of:

- A. Failed fusion between maxillary & mandibular processes
- B. Failed fusion between the two lateral nasal folds
- C. Necrosis of neural crest

others

4. The cranial nerve supply muscles develop from the second pharyngeal arch is:

- A. Glossopharyngeal
- B. Trigeminal nerve
- C. Oculomotor nerve
- D. Facial nerve
- E. Vagus nerve

5. Cleft lip is caused by failure of:

- A. Fusion of lateral nasal prominence and maxillary process
- B. Fusion of medial nasal prominence and maxillary process
- C. Fusion of medial nasal prominence and lateral nasal prominence
- D. Fusion of lateral nasal prominence and mandibular process
- E. Fusion of medial nasal prominence and mandibular process

6. is NOT a derivative of the 1st pharyngeal arch:

- A. Sphenomandibular ligament
- B. Anterior belly of digastric muscle
- C. Malleus
- D. Parathyroid gland
- E. Temporalis

7. Tensor tympani is derived from:

- A. Fifth pharyngeal arch
- B. First pharyngeal arch
- C. Third pharyngeal arch
- D. Fourth pharyngeal arch
- E. Second pharyngeal arch

8. The Reichert cartilage develops into:

- A. Malleus
- B. Nasal cavity
- C. Mandible
- D. Stapes
- E. Muscles of mastication

9. The cranial nerve supply muscles develop from the first pharyngeal arch is:

- A. Oculomotor nerve
- B. Glossopharyngeal nerve
- C. Trigeminal nerve
- D. Vagus nerve
- E. Facial nerve

10. External auditory meatus develops from :

- A. Second pharyngeal arch
- B. Second pharyngeal cleft
- C. First pharyngeal cleft
- D. Third pharyngeal pouch
- E. Second pharyngeal pouch

11. Greater horn and lower part of the body of hyoid bone are derived from the:

- A. Third pharyngeal arch
- B. First pharyngeal arch
- C. Occipital myotomes
- D. Fourth pharyngeal arch
- E. Preotic myotomes

12. The philtrum of the upper lip is formed of which of the following processes?

- a. Maxillary
- b. Mandibular
- c. Intermaxillary segment
- d. Lateral nasal

13. Which of the following structures is involved in the formation of the cleft palate?

- a. Second pharyngeal arch
- b. Third pharyngeal arch
- c. The palatine plates
- d. The maxillary and medial nasal processes

14. Which of the following structures is a derivative of the second pharyngeal arch?

- a. Temporalis
- b. Masseter
- c. Lateral pterygoid
- d. Medial pterygoid
- e. Occipitofrontalis

15. The 1st branchial cleft gives rise to which of the following structures?

- a. Nasal cavity
- b. Tympanic cavity
- c. Oral cavity
- d. External acoustic meatus
- e. Auditory tube

16. Which of the following structures is a derivative of the second pharyngeal arch?

- a. Malleus
- b. Mandible
- c. Sphenomandibular ligament
- d. Incus
- e. Styloid process

17. Thymus gland is a derivative of which of the following pouches?

- a. 1st
- b. 2nd
- c. 3rd
- d. 4th

Answers:

1-A | 2-B | 3-A | 4-D | 5-B | 6-D | 7-B | 8-D | 9-C |
10-C | 11-A | 12-C | 13-C | 14-E | 15-D | 16-E | 17-C

Formatives

1. The superior orbital fissure is situated between:

- a. Medial wall and roof of orbit
- b. Medial wall and floor of orbit
- c. Lateral wall and roof of orbit
- d. Lateral wall and floor of orbit

2. The nerve supply of medial rectus muscle is:

- a. Oculomotor
- b. Trochlear
- c. Abducent
- d. Ophthalmic

3. Ophthalmic artery enters orbit through:

- a. optic canal
- b. superior orbital fissure
- c. Inferior orbital fissure
- d. Infraorbital foramen

4. The parasympathetic nerve supply of the lacrimal apparatus comes from:

- a. Oculomotor nerve
- b. Facial nerve
- c. Glossopharyngeal nerve
- d. Vagus nerve

5. The branch of ophthalmic artery which supplies the nasal cavity is:

- a. Lacrimal artery
- b. Central retinal artery
- c. Anterior and posterior ethmoidal branches
- d. Supra-orbital artery

6. Which muscle is responsible for adduction and lateral rotation of eyeball?

- a. Superior rectus
- b. Inferior rectus
- c. Medial rectus
- d. Lateral rectus

7. Which structure forms the innermost layer of eye?

- a. Sclera
- b. Choroid
- c. Retina
- d. Lens

8. Orbital plate of maxillary bone forms:

- a. Roof of the orbit
- b. Floor of the orbit
- c. Medial wall of the orbit
- d. Lateral wall of the orbit

9. The extra-ocular muscle which is supplied by the trochlear nerve:

- a. Superior rectus
- b. Inferior rectus
- c. Inferior oblique
- d. Superior oblique

10. Ciliary ganglion is a site of relay of parasympathetic fibers that supply:

- a. Lacrimal gland
- b. Parotid gland
- c. Sphincter pupillae muscle
- d. Submandibular gland

11. The parasympathetic supply of the lacrimal gland is derived from:

- a. Ciliary ganglion
- b. Pterygo-palatine (Spheno-palatine) ganglion
- c. Otic ganglion
- d. Submandibular ganglion

Previous Exams

12. Floor of the orbit formed by:

- a. Frontal bone
- b. Zygomatic bone
- c. Sphenoid bone
- d. Nasal bone

13. Which divisions of ophthalmic artery supply the nasal cavity:

- a. Supraorbital & supratrochlear a.
- b. Centro retinal & ciliary a.
- c. Anterior & posterior ethmoidal a.
- d. Lacrimal artery

14. Which divisions of ophthalmic artery supply the scalp & face:

- a. Supraorbital & supratrochlear a.
- b. Centro retinal & ciliary a.
- c. Anterior & posterior ethmoidal a.
- d. Lacrimal artery

15. It's responsible for high acuity vision:

- a. Fovea centralis
- b. Blind spot
- c. Pigmented outer layer of retina
- d. Macula lutea

16. Zygomatic bone share in formation of:

- a. Roof and Latral wall of the orbit
- b. Roof and medial wall of the orbit
- c. Floor and Latral wall of the orbit
- d. Floor and medial wall of the orbit

18. Superior & inferior ophthalmic veins drains onto:

- a. Superior sagittal sinus
- b. Inferior sagittal sinus
- c. Cavernous sinuse
- d. Sigmoid sinus

Answers :

1-c | 2-a | 3-a | 4-b | 5-c | 6-b | 7-c | 8-b | 9-d | 10-c | 11-b | 12-b | 13-c | 14-a | 15-a | 16-c | 18-c

Formatives

1. The most depressed area of the tympanic membrane corresponds to the tip of the manubrium of handle of malleus.

- a- Pars tensa
- b- Pars flaccida
- c- Umbo
- d- Cone of light reflection

2. A thin plate of bone separating the middle ear cavity from the middle cranial fossa

- a- Tegmen Tympani
- b- Mastoid antrum
- c- Pyramid
- d- Processus cochleariformis

3. The promontory of the 1st coil of cochlea forms...

- a- Roof
- b- Floor
- c- Medial wall
- d- Lateral wall

4. A sensory pain supply of tympanic cavity is mediated by

- a- Auriculo-temporal nerve
- b- Great auricular nerve
- c- Facial nerve
- d- Tympanic plexus of glossopharyngeal nerve

5. They are considered for detection of head rotation:

- a- Utricle
- b- Sacculle
- c- Semicircular ducts
- d- Cochlear duct

6. The central part of the bony labyrinth is

- a- cochlea
- b- vestibule
- c- semicircular duct
- d- utricle

7. Most medial bone of middle ear is

- a- Malleus
- b- Incus
- c- Stapes
- d- Hammer

Previous exams

8. Lower canal of the anterior wall of middle ear is the :

- a- Canal of tensor tympani ms.
- b- Canal of tensor palati ms.
- c- Eustachian tube
- d- Canal for optic nerve

9. It forms the Medial wall of tympanic cavity :

- a- Tympanic membrane

b- Tegmen tympani

c- Vertical part of bony facial canal

d- Promontory of 1st turn of cochlea

10. which inner layer structure is connected to the stapes of the middle ear :

- a- Oval window
- b- Cochlear duct
- c- Round windows
- d- Auditory (pharyngotympanic) tube

11. the inner ear is located within :

- a- Squamous temporal bone
- b- Tympanic temporal bone
- c- Petrous temporal bone
- d- Mastoid bone

Answers:

1-C | 2-A | 3-C | 4-D | 5-C | 6-B | 7-C | 8-C | 9-D | 10-A | 11-C

1. What is the only part of the auricle (pinna) to NOT be supported by elastic cartilage?

- A. Helix
- B. Antihelix
- C. Lobule
- D. Tragus

2. Which nerve supplies sensory innervation to the skin of the ear canal (external acoustic meatus) on its anterior wall and roof?

- A. Auricular branch of vagus
- B. Chorda tympani
- C. Auriculotemporal nerve
- D. Facial nerve

3. What is the name of the inward convexity on the inner surface of the tympanic membrane where the handle of the malleus attaches?

- A. Pars flaccida
- B. Cone of light
- C. Umbo
- D. Promontory

4. Which part of the middle ear cavity lodges the head of the malleus and the body of the incus?

- A. Tympanic sinus
- B. Mastoid antrum
- C. Auditory tube
- D. Epitympanic recess

5. Which structure forms the ROOF of the middle ear cavity, separating it from the middle cranial fossa?

- A. Jugular plate
- B. Tegmen tympani
- C. Bony facial canal
- D. Promontory

6. The stapedius muscle, which dampens loud sounds, is innervated by which cranial nerve?

- A. Trigeminal (V)
- B. Facial (VII)
- C. Glossopharyngeal (IX)
- D. Vagus (X)

7. What is the primary function of the auditory (Eustachian) tube?

- A. Amplify sound vibrations
- B. Equalize air pressure on both sides of the tympanic membrane
- C. Transmit sound waves to the cochlea
- D. Drain fluid from the middle ear

8. Which structure of the inner ear is directly responsible for hearing?

- A. Semicircular canals
- B. Utricle
- C. Sacculle
- D. Cochlea

9. What is the fluid called that fills the space BETWEEN the bony and membranous labyrinths?

- A. Endolymph
- B. Perilymph
- C. Cerebrospinal fluid
- D. Blood plasma

10. Which structure of the bony labyrinth contains the sensory receptors (maculae) for detecting linear head acceleration and static equilibrium?

- A. Cochlea
- B. Semicircular canals
- C. Vestibule
- D. Modiolus

11. Which window of the inner ear is closed by the footplate of the stapes?

- A. Round window
- B. Oval window
- C. Cochlear window
- D. Vestibular window

12. The spiral organ of Corti, the actual hearing receptor, is located within which structure?

- A. Scala vestibuli
- B. Scala tympani
- C. Cochlear duct
- D. Sacculle

13. Which chamber of the cochlea lies directly BELOW the cochlear duct?

- A. Scala media
- B. Scala vestibuli
- C. Scala tympani
- D. Utricle

14. Which artery is a major source of blood supply to the structures of the inner ear (labyrinth)?

- A. Superficial temporal artery
- B. Middle meningeal artery
- C. Labyrinthine artery
- D. Deep auricular artery

15. The tensor tympani muscle, which tenses the tympanic membrane, is innervated by a branch of which nerve?

- A. Facial nerve
- B. Glossopharyngeal nerve
- C. Mandibular nerve (V3)
- D. Vagus nerve

16. Which structure is found on the MEDIAL wall of the middle ear and represents the basal turn of the cochlea?

- A. Oval window
- B. Pyramid
- C. Promontory
- D. Processus cochleariformis

17. The sensory receptor (crista ampullaris) for detecting rotational head movements is located in the ampullae of which structures?

- A. Cochlear ducts
- B. Sacculle and utricle
- C. Semicircular ducts
- D. Scala vestibuli and tympani

18. What is the name of the thin membrane that closes the round window?

- A. Primary tympanic membrane
- B. Secondary tympanic membrane
- C. Vestibular membrane
- D. Basilar membrane

19. Which nerve carries sensory information from the mucosa lining the inner surface of the tympanic membrane and the middle ear cavity?

- A. Tympanic plexus (from CN IX)
- B. Chorda tympani (from CN VII)
- C. Auriculotemporal nerve (from V3)
- D. Auricular branch of vagus (CN X)

20. During an otoscopic examination, the "cone of light" is a reflection seen on the tympanic membrane. Where is it normally located?

- A. Anterosuperior quadrant
- B. Anteroinferior quadrant
- C. Posterosuperior quadrant
- D. Posteroinferior quadrant

Answers:

1-C | 2-C | 3-C | 4-D | 5-B | 6-B | 7-B | 8-D | 9-B | 10-C | 11-B | 12-C | 13-C | 14-C | 15-C | 16-C | 17-C | 18-B | 19-A | 20-B

Formatives

1. Inferior quadrigeminal brachium conveys fibers from:

- a. Inferior colliculus to the medial geniculate body
- b. Superior colliculus to the lateral geniculate body
- c. Inferior colliculus to the lateral geniculate body
- d. Superior colliculus to the medial geniculate body

2. The facial colliculus is a bulging formed by:

- a. Facial nucleus and abducent nerve
- b. Facial fibers and abducent nucleus
- c. Facial fibers and facial nucleus
- d. Abducent fibers and facial nucleus

3. Cranial nerve IV (Trochlear) is located inferior to which landmark?

- a. Superior colliculus
- b. Inferior colliculus
- c. Pons
- d. Thalamus

4. The posterior white matter of medulla oblongata medially contains:

- a. Fasciculus gracilis
- b. Fasciculus cuneatus
- c. Spinal tract of trigeminal nerve
- d. Internal arcuate fibers

5. These nerves are seen in the ponto-cerebellar angle:

- a. Oculomotor & Trochlear nerves
- b. Trigeminal & Abducent nerves
- c. Facial & Vestibulocochlear nerves
- d. Glossopharyngeal & Vagus nerves

6. A diamond-shaped space in the base of the forebrain, bounded by the optic chiasm and optic tract anteriorly & the cerebral peduncles posteriorly:

- a. Fourth ventricle
- b. Interpeduncular fossa
- c. Foramen magnum
- d. Tentorium cerebelli

7. Blackish coloration due to its high content of polymerized dopamine:

- a. Red nucleus
- b. Pontine nuclei
- c. Substantia nigra
- d. Inferior olivary nucleus

8. Cells of the vestibular ganglion project their central processes to:

- a. Cerebellum & vestibular nuclei
- b. Cerebellum & cochlear nucleus
- c. Cerebellum & main sensory nucleus

d. Cerebrum & posterior horn cells of the spinal cord

9. On physiological basis, the hyperacusis associated with Bell's palsy is due to paralysis of:

- a. Stapedius muscle
- b. Tensor tympani muscle
- c. Tensor palati muscle
- d. Both tensor tympani and tensor palati muscles

10. Input fibers to the cerebellar cortex that end directly on Purkinje cells:

- a. Mossy fibers
- b. Olivo-cerebellar fibers
- c. Pontocerebellar fibers
- d. Cuneocerebellar fibers

Exams

11. Tumor in cerebellopontine angle affects:

- a. Oculomotor nerve
- b. Trigeminal nerve
- c. Abducent nerve
- d. Facial nerve

12. Fibers from nucleus gracilis and cuneatus decussate and form:

- a. Medial lemniscus
- b. Lateral lemniscus
- c. Spinal lemniscus
- d. Trigeminal lemniscus

13. Cranial nerves visible on surface of medulla oblongata are:

- a. Three cranial nerves
- b. Four cranial nerves
- c. Five cranial nerves
- d. Six cranial nerves

14. Gracile nucleus has the following functions:

- a. unconscious proprioception from the upper limb
- b. Proprioception from the lower half of the body
- c. Proprioception from the upper half of the body
- d. Proprioception from the face

15. Nucleus for motor functions to pharynx, larynx & soft palate :

- a. Cuneate nucleus
- b. inferior olivary nucleus
- c. hypoglossal nucleus
- d. nucleus ambiguus

Answers:

1-a | 2-b | 3-b | 4-a | 5-c | 6-b | 7-c | 8-a | 9-a | 10-b | 11-d | 12-a | 13-c | 14-b | 15.d

1. Which is the widest and most superior segment of the brainstem?

- a) Medulla oblongata
- b) Pons
- c) Midbrain
- d) Cerebellum

2. What part of the brainstem is located between the midbrain superiorly and the medulla inferiorly?

- a) Pons
- b) Cerebral peduncle
- c) Tectum
- d) Medial lemniscus

3. The medulla oblongata extends from which structure inferiorly?

- a) Inferior pontine sulcus
- b) Decussation of the great pyramids
- c) Interpeduncular fossa
- d) Cerebellar peduncles

4. Which cranial nerve emerges from the anterolateral sulcus of the medulla?

- a) Hypoglossal nerve
- b) Vagus nerve
- c) Accessory nerve
- d) Glossopharyngeal nerve

5. What forms the pyramid of the medulla oblongata?

- a) Corticospinal tract
- b) Gracile nucleus
- c) Cuneate nucleus
- d) Inferior olivary nucleus

6. Which structure connects the medulla to the cerebellum?

- a) Superior cerebellar peduncle
- b) Middle cerebellar peduncle
- c) Inferior cerebellar peduncle
- d) Olive

7. What separates the back of the medulla from the back of the pons?

- a) Decussation of pyramids
- b) Medullary striae of the 4th ventricle
- c) Dorsal column nuclei
- d) Inferior fovea

8. The gracile tubercle in the medulla is formed by which nucleus?

- a) Inferior olivary nucleus
- b) Gracile nucleus
- c) Cuneate nucleus
- d) Hypoglossal nucleus

9. Which of the following cranial nerve nuclei is responsible for taste sensation?

- a) Nucleus ambiguus
- b) Dorsal motor nucleus of vagus
- c) Nucleus solitarius
- d) Hypoglossal nucleus

10. What is the location of the inferior olivary nucleus in the medulla?

- a) Lateral to the pyramid
- b) Medial to the gracile nucleus
- c) Beneath the 4th ventricle
- d) Posterior to the cuneate tubercle

11. What is the function of the hypoglossal nerve?

- a) Supplies the muscles of the pharynx
- b) Supplies the muscles of the tongue
- c) Supplies the parotid gland
- d) Supplies the laryngeal muscles

12. Which nucleus provides motor fibers to the glossopharyngeal nerve?

- a) Dorsal motor nucleus of vagus
- b) Inferior salivatory nucleus
- c) Nucleus ambiguus
- d) Spinal nucleus of trigeminal

13. Which part of the medulla forms the lower part of the floor of the 4th ventricle?

- a) Closed medulla
- b) Open medulla
- c) Pyramids
- d) Olives

14. Which nerve carries proprioception from the muscles of mastication?

- a) Facial nerve
- b) Glossopharyngeal nerve
- c) Trigeminal nerve
- d) Abducent nerve

15. Where is the nucleus ambiguus located?

- a) Pons
- b) Medulla oblongata
- c) Midbrain
- d) Cerebellum

16. Which cranial nerve emerges between the pons and pyramid of the medulla?

- a) Trigeminal nerve
- b) Abducent nerve
- c) Facial nerve
- d) Vestibulocochlear nerve

17. What does the medial lemniscus carry?

- a) Proprioception and fine touch
- b) Pain and temperature
- c) Crude touch
- d) Hearing sensation

18. The corticospinal tract forms which structure in the medulla?

- a) Gracile tubercle
- b) Pyramid
- c) Olive
- d) Inferior cerebellar peduncle

19. Which cranial nerve nucleus is responsible for parasympathetic supply to the parotid gland?

- a) Dorsal motor nucleus of vagus
- b) Inferior salivatory nucleus
- c) Nucleus ambiguus
- d) Edinger-Westphal nucleus

20. Which structure forms the facial colliculus?

- a) Abducent nucleus
- b) Facial nerve root
- c) Both a and b
- d) None of the above

21. Which is the auditory reflex center?

- a) Superior colliculus
- b) Inferior colliculus
- c) Red nucleus
- d) Substantia nigra

22. Which cranial nerve exits at the cerebellopontine angle?

- a) Oculomotor nerve
- b) Abducent nerve
- c) Facial nerve
- d) Glossopharyngeal nerve

23. The trochlear nerve supplies which muscle?

- a) Superior oblique
- b) Inferior oblique
- c) Lateral rectus
- d) Medial rectus

24. The tectum of the midbrain contains which structures?

- a) Red nucleus
- b) Superior and inferior colliculi
- c) Crus cerebri
- d) Substantia nigra

25. What is the function of the red nucleus?

- a) Reflex auditory center
- b) Facilitates tone of flexor muscles
- c) Reflex movements of the eyes
- d) Secretes dopamine

26. What separates the pons from the open medulla?

- a) Medullary striae of the 4th ventricle
- b) Basilar groove
- c) Inferior fovea
- d) Superior cerebellar peduncles

27. Which fibers form the trapezoid body in the brainstem?

- a) Ventral cochlear nucleus fibers
- b) Dorsal cochlear nucleus fibers
- c) Substantia nigra fibers
- d) Olivocerebellar fibers

28. What does the lateral lemniscus carry?

- a) Proprioception sensation
- b) Hearing sensation
- c) Pain and temperature
- d) Crude touch

29. Which nucleus receives fibers from the organ of Corti?

- a) Superior olivary nucleus
- b) Dorsal cochlear nucleus
- c) Nucleus ambiguus
- d) Trigeminal nucleus

30. Which cranial nerve emerges from the interpeduncular fossa?

- a) Oculomotor nerve
- b) Trochlear nerve
- c) Facial nerve
- d) Abducent nerve

Horizontal Answers:

1-c | 2-a | 3-b | 4-a | 5-a | 6-c | 7-b | 8-b | 9-c | 10-a |
11-b | 12-c | 13-b | 14-c | 15-b | 16-b | 17-a | 18-b |
19-b | 20-c | 21-b | 22-c | 23-a | 24-b | 25-b | 26-a |
27-a | 28-b | 29-b | 30-a

Formatives

1- The blood brain barrier is....

- a- A barrier between blood and CSF and Prevents the passage of some substances, such as antibiotics and chemicals and bacterial toxic matter, from blood to the nerve tissue.
- b- Contains thick and non-continuous BM
- c- Formed of tight junctions between fenestrated endothelium
- d- Astrocyte End-Feet are inside the basement membrane.

2. The site of contact between two neurons is called

- a- dendrites
- b- Axons
- c- Synapses
- d- Cell body

3- When drug molecules pass from the blood to the neuron which of the following cells will control their passage?

- a- Oligodendrocyte.
- b- Microglia.
- c- Astrocytes.
- d- Schwann cells

4. The synaptic knob contains:

- a- Mitochondria only to supply energy.
- b- Mitochondria to supply energy for synthesis of and vesicles containing the chemical transmitter.
- c- Receptors of the chemical transmitter.
- d- Enzymes to hydrolyze the chemical transmitter.

5- Which of the following cells are responsible for Myelination of NS nerve fibers?-

- a- Microglia
- b- Oligodendrocytes
- c- Schwann cells
- d- Ependymal cells

6- Which of the followings are the phagocytic cells of the brain?

- a- Microglia
- b- Oligodendrocytes
- c- Schwann cells
- d- Ependymal cells

7- What is the structure of the pia mater?

- a- It is formed of avascular Fibro-elastic connective tissue
- b- It is formed of delicate vascular connective tissue
- c- It is formed of thick white fibrous connective tissue
- d- It is formed of web like strands of collagen fibers

8- What is meant by the point of Connection between one neuron to another?

- a- Ganglia
- b- Neuroglia
- c- Synapse
- d- Nucleu

Answers :

1-a | 2-c | 3-c | 4-b | 5-b | 6-a | 7-b | 8-c

Doctor Questions

1. The functional unit of the nervous system is the:

- A. Neuroglia
- B. Neuron
- C. Axon
- D. Synapse

2. Connective tissue-like supporting cells in the CNS are called:

- A. Schwann cells
- B. Satellite cells
- C. Neuroglia
- D. Microglia

3. Melanin pigment is normally found in the:

- A. Cerebellum
- B. Substantia nigra
- C. Pons
- D. Medulla

4. All of the following are true about dendrites EXCEPT:

- A. Numerous and short
- B. Non-myelinated
- C. Carry impulses away from cell body
- D. Have Nissl bodies

5. Axons differ from dendrites by:

- A. Being usually one per cell
- B. Containing Nissl bodies
- C. Being short and numerous
- D. Receiving impulses only

6. Pseudo-unipolar neurons are characteristic of:

- A. Autonomic ganglia
- B. Spinal (dorsal root) ganglia
- C. Sympathetic chain
- D. Brain stem nuclei

7. Multipolar neurons are typically found in:

- A. DRG
- B. Retina
- C. Autonomic ganglia
- D. Olfactory mucosa

8. Myelination in the CNS is done by:

- A. Schwann cells
- B. Oligodendrocytes
- C. Astrocytes
- D. Microglia

9. Astrocytes are mainly associated with formation of:

- A. Myelin sheath
- B. CSF
- C. BBB
- D. Nodes of Ranvier

10. According to number of processes, a pseudo-unipolar neuron has:

- A. 1 process
- B. 2 processes
- C. Many processes
- D. No processes

11. Axo-axonic synapse means a synapse between:

- A. Axon and soma
- B. Axon and dendrite
- C. Axon and axon
- D. Dendrite and dendrite

12. The type of synapse that uses neurotransmitters is:

- A. Chemical
- B. Electrical
- C. Mixed
- D. Direct

13. A ganglion is defined as:

- A. Cell bodies inside CNS
- B. Nerve fibers inside CNS
- C. Aggregation of nerve cell bodies outside CNS
- D. Aggregation of axons outside CNS

14. The meninges arranged from outer to inner are:

- A. Dura → Pia → Arachnoid
- B. Pia → Arachnoid → Dura
- C. Dura → Arachnoid → Pia
- D. Arachnoid → Dura → Pia

15. The subarachnoid space contains:

- A. Blood only
- B. CSF
- C. Myelin
- D. Lymph

16. Choroid plexus is formed by:

- A. Flattened arachnoid
- B. Invaginated pia mater
- C. Astrocytes
- D. Satellite cells

17. Capillaries in the choroid plexus are:

- A. Non-fenestrated
- B. Fenestrated
- C. Continuous tight junctions
- D. Absent

18. All of the following are brain barriers EXCEPT:

- A. Blood–brain barrier
- B. Blood–CSF barrier
- C. Brain–CSF barrier
- D. Myelin–CSF barrier

19. In the blood–brain barrier (BBB), capillaries are:

- A. Fenestrated
- B. Non-fenestrated
- C. Discontinuous
- D. Absent

20. In the blood–CSF barrier, capillaries are:

- A. Non-fenestrated
- B. Fenestrated
- C. Sinusoidal
- D. Absent

Answers :

1-B | 2-C | 3-B | 4-C | 5-A | 6-B | 7-C | 8-B | 9-C |
10-A | 11-C | 12-A | 13-C | 14-C | 15-B | 16-B |
17-B | 18-D | 19-B | 20-B

Formatives

1. Betz cells are seen in:

- A. Molecular layer
- B. Inner pyramidal
- C. Outer pyramidal
- D. Multiform layer

2. Stellate cells of cerebellum present in:

- A. Granular
- B. Purkinje
- C. Multiform
- D. Molecular layer

3. Which of the following form myelin in the peripheral nerve

- A. Schwan cell
- B. Oligodendrocyte
- C. Satellite cells
- D. Microglia

4. Which sensory receptor in the skin is without a capsule

- A. Kruse end bulb
- B. Pacinian corpuscle
- C. Merkel's disc
- D. Ruffini ending

5. Pacinian corpuscles usually are considered to be most responsive to

- A. light touch
- B. pressure
- C. pain
- D. heat.

6. A 30-year-old man was severed from impaired movement, fatigue, vision troubles and he diagnosed as having a demyelinating disease or multiple sclerosis..... cells are responsible for the myelin synthesis in the CNS and.....in the PNS respectively

- a. Schwann cells/ oligodendroglia cells
- b. Microglia cells / oligodendroglia cells
- c. oligodendroglia cells / Microglia cells
- d. oligodendroglia cells/ Schwann cells

Answers :

1-b | 2-d | 3-a | 4-c | 5-b | 6-d

1. Pyramidal cells are characteristic of the

- a. Ventral horn of spinal cord
- b. Dorsal horn of spinal cord
- c. Cerebral cortex
- d. Cerebellar cortex

2. The largest neurons in the central nervous system (pyramidal or Betz Cells) are found in which layer of cerebral cortex:

- a. Layer II
- b. Layer V

c. Layer III

d. Layer IV

3. The cerebellar cortex contains huge neurons that have fine dendrites and axons named as

- a. Basket cell layer
- b. Stellate cell layer
- c. Purkinje cell layer
- d. Granule cell layer

4. White matter is mainly composed of

- a. Myelinated axons
- b. Unmyelinated axons
- c. Unmyelinated dendrites
- d. Unmyelinated nerve cell bodies

5. In the spinal cord:

- a. Gray matter sits internal to the white matter
- b. White mater sits internal to the grey matter
- c. There is no grey matter
- d. White matter and grey matter are mixed into a homogenous mixture

6. Giant pyramidal cells in the primary motor cortex of the frontal lobe called..

- a. Inner pyramidal cells
- b. Outer pyramidal cells
- c. Betz cells
- d. Purkinje cells

7. The white matter of the spinal cord contain..

- a. Unmyelinated axons and Oligo-dendrocytes
- b. Myelinated axons and Schwann cells
- c. Unmyelinated axons and Schwann cells
- d. Myelinated axons and Oligo-dendrocytes

8. The inner pyramidal layer of the cerebral cortex formed of..

- a. Medium sized pyramidal cells
- b. large pyramidal cells
- c. Small pyramidal cells
- d. Betz cells

9. The innermost layer of the cerebellar cortex is the....

- a. inner pyramidal layer
- b. outer pyramidal layer
- c. granular layer
- d. purkinje layer

10. The oblique interruptions of myelin sheath called....

- a. Nodes of Ranvier
- b. Axon terminal
- c. Lantermann clefts
- d. Schwann cells clefts

11. Which of the following cells is responsible for myelination of nerve fibers in CNS?

- A. Schwann cell
- B. Oligodendrocyte
- C. Astrocyte
- D. Ependymal cell

12. Cerebral cortex is organized into.....layers

- a. six horizontal layers
- b. three horizontal layers
- c. six vertical layers
- d. three vertical layers

13. Meissner's corpuscles:

- a. Is surrounded by CT capsule responsible for touch sensation
- b. Is an unencapsulated receptor responsible for cold sensation
- c. Is an unencapsulated receptor responsible for touch sensation
- d. Is surrounded by CT capsule response for hot sensation

Answers :

1-c | 2-b | 3-c | 4-a | 5-a | 6-c | 7-d | 8-b | 9-c | 10-c |
11-b | 12-a | 13-a

Formatives

1. The CBF in an adult is:

- A. Not affected by variation of the ICP.
- B. Not affected by the PaCO₂ or hypoxia.
- C. 750 milliliters per minute, or about 15% of cardiac output.
- D. 75 ml per 100g per min.

2. CBF =

- A. CPP / CVR.
- B. MAP – ICP.
- C. COP x HR.
- D. 150 mL/min.

3. The BBB:

- A. Allows the passage of many substances and drugs from the blood to the brain.
- B. Is due to the high permeability of cerebral capillaries.
- C. Limits the passage of many substances and drugs from the blood to the brain

4. Lumbar Puncture:

- A. It involves passing a needle into the subarachnoid space at the levels above L2 vertebra.
- B. Is a sterile procedure, done to obtain CSF samples for diagnostic purposes.
- C. Is safe and has no hazards.
- D. Is always not followed by headache.

5. The CBF in an adult is:

- A. Not affected by variation of the ICP.
- B. Not affected by the PaCO₂ or hypoxia.
- C. 750 milliliters per minute, or about 15% of cardiac output.
- D. 75 ml per 100g per min.

6. CBF =

- A. CPP / CVR.
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- C. COP x HR.
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7. Lumbar Puncture:

- A. It involves passing a needle into the subarachnoid space at the levels above L2 vertebra.
- B. Is a sterile procedure, done to obtain CSF samples for diagnostic purposes.
- C. Is safe and has no hazards.
- D. Is always not followed by headache.

8. The blood brain barrier is....

- A. A barrier between blood and CSF and Prevents the passage of some substances, such as antibiotics and chemicals and bacterial toxic matter, from blood to the nerve tissue.
- B. Contains thick and non-continuous BM

C. Formed of tight junctions between fenestrated endothelium

D. Astrocyte End-Feet are inside the basement membrane.

Answer:

1-C | 2-A | 3-C | 4-B | 5-C | 6-A | 7-B | 8-A

Previous Exams

1. The BBB:

- a. Allows the passage of many substances and drugs from the blood to the brain.
- b. Is due to the high permeability of cerebral capillaries.
- c. Limits the passage of many substances and drugs from the blood to the brain.
- d. Is an absolute barrier.

2. Lumbar Puncture:

- a. It involves passing a needle into the subarachnoid space at the level of the L2 vertebra.
- b. Is a sterile procedure, done to obtain CSF samples for diagnostic purposes.
- c. Is safe and has no hazards.
- d. Is always not followed by headache.

3. The CSF is absorbed at the:

- a. Superior sagittal sinus.
- b. Lateral ventricle.
- c. Third ventricle.
- d. Inferior sagittal sinus.

4. prevents the passage of some substances, such as antibiotics, chemicals, and bacterial toxins, from blood to the nerve tissue:

- a. Brain CSF Barrier.
- b. Blood CSF Barrier.
- c. Blood Brain Barrier.
- d. The dura mater.

5. Cerebral arterioles dilate primarily in response to:

- a. Low levels of carbon dioxide
- b. High levels of carbon dioxide
- c. High oxygen levels
- d. Sympathetic stimulation

6. The typical cerebral blood flow (CBF) in an adult human is approximately:

- A. 250 ml/min
- B. 500 ml/min
- C. 750 ml/min
- D. 1200 ml/min

7. ICP of 20 mmHg is considered:

- A. Normal
- B. Low intracranial pressure

- C. Intracranial hypotension
- D. Intracranial hypertension

Answers: 1-C | 2-B | 3-A | 4-C | 5-B | 6-C | 7-D

Dr. Selim Questions

1. The CBF (Cerebral Blood Flow) in an adult is:

- A. Not affected by variation of the ICP
- B. Not affected by the PaCO_2 or hypoxia
- C. 750 milliliters per minute, or about 15% of cardiac output
- D. 75 ml per 100g per min

2. CBF =

- A. CPP / CVR
- B. $\text{MAP} - \text{ICP}$
- C. $\text{COP} \times \text{HR}$
- D. 150 mL/min

3. The BBB (Blood-Brain Barrier):

- A. Allows the passage of many substances and drugs from the blood to the brain
- B. Is due to the high permeability of cerebral capillaries
- C. Limits the passage of many substances and drugs from the blood to the brain
- D. None of the above

4. Lumbar Puncture:

- A. It involves passing a needle into the subarachnoid space at the levels above L2 vertebra
- B. Is a sterile procedure, done to obtain CSF samples for diagnostic purposes
- C. Is safe and has no hazards
- D. Is always not followed by headache

5. The blood-brain barrier is:

- A. A barrier between blood and CSF, blocking bacterial toxic matter from blood to the nerve tissue
- B. Prevents the passage of some substances, such as antibiotics and chemicals
- C. Contains thick and non-continuous BM
- D. Formed of tight junctions between fenestrated endothelium

Answers: 1-C | 2-A | 3-C | 4-B | 5-B

Dr. Wagdy Questions

1. The typical cerebral blood flow (CBF) in an adult human is approximately:

- A. 250 ml/min
- B. 500 ml/min
- C. 750 ml/min
- D. 1200 ml/min

2. Cerebral blood flow represents roughly what percentage of the cardiac output?

- A. 5%
- B. 10%
- C. 15%
- D. 25%

3. Average brain perfusion is approximately:

- A. 20–30 ml/100g/min
- B. 50–54 ml/100g/min
- C. 80–100 ml/100g/min
- D. 10–15 ml/100g/min

4. Increased CBF (hyperemia) can lead primarily to:

- A. Decreased ICP
- B. Increased ICP
- C. Increased CPP
- D. Hypotension

5. Cerebral ischemia occurs when CBF drops below:

- A. 50 ml/100g/min
- B. 30 ml/100g/min
- C. 18–20 ml/100g/min
- D. 60 ml/100g/min

6. Brain tissue death occurs when CBF falls below:

- A. 15 ml/100g/min
- B. 12–14 ml/100g/min
- C. 8–10 ml/100g/min
- D. 20 ml/100g/min

7. The ischemic cascade is triggered when:

- A. ICP increases
- B. PaCO_2 falls
- C. Brain tissue becomes ischemic
- D. CPP rises above normal

8. The normal value of cerebral perfusion pressure (CPP) should be above:

- A. 20 mmHg
- B. 30 mmHg
- C. 40 mmHg
- D. 50 mmHg

9. Intracranial pressure (ICP) should normally NOT exceed:

- A. 5 mmHg
- B. 10 mmHg
- C. 15 mmHg
- D. 25 mmHg

10. ICP of 20 mmHg is considered:

- A. Normal
- B. Low intracranial pressure
- C. Intracranial hypotension
- D. Intracranial hypertension

11. Each 1 mmHg increase in PaCO₂ (between 20–60 mmHg) changes CBF by approximately:

- A. 0.5 ml/100g/min
- B. 1–2 ml/100g/min
- C. 3–5 ml/100g/min
- D. No effect

12. Cerebral ischemia caused by high ICP can trigger which reflex?

- A. Babinski reflex
- B. Baroreceptor reflex
- C. Cushing's reflex
- D. Diving reflex

13. Regional CBF (rCBF) at a specific location can be measured over time in the operating room using:

- A. PET
- B. Thermal diffusion
- C. Ultrasound
- D. CT

14. Lateral ventricle CSF flows to third ventricle via:

- A. Aqueduct of Sylvius
- B. Foramen of Monro
- C. Foramina of Luschka
- D. Foramen of Magendie

15. Third ventricle connects to fourth ventricle via:

- A. Foramen of Monro
- B. Cerebral aqueduct
- C. Foramina of Luschka
- D. Choroid plexus

16. CSF flows to subarachnoid space via:

- A. Foramina of Luschka and Magendie
- B. Aqueduct only
- C. Arachnoid villi
- D. Choroid plexus

17. Syringomyelia results from:

- A. Blockage of CSF circulation
- B. Hydrocephalus
- C. ICP increase
- D. Brain tumor

18. Most common Lumbar Puncture level:

- A. L2–L3
- B. L3–L4
- C. L4–L5
- D. L5–S1

Answers :

1-C | 2-C | 3-B | 4-B | 5-C | 6-C | 7-C | 8-D | 9-C |
10-D | 11-B | 12-C | 13-B | 14-B | 15-B | 16-A |
17-A | 18-

Formative

1. Once neurotransmitter is released, it typically produces signals in postsynaptic neurons by:

- Entering the postsynaptic neuron
- Attaching to vesicles
- Binding to presynaptic receptors
- Binding to postsynaptic receptors

2. The first chemical to be identified as a neurotransmitter was:

- Norepinephrine
- Epinephrine
- Dopamine
- Acetylcholine

3. The BBB (Blood-Brain Barrier):

- Allows the passage of many substances and drugs from the blood to the brain
- Is due to the high permeability of cerebral capillaries
- Limits the passage of many substances and drugs from the blood to the brain

4. The synaptic knob contains:

- Mitochondria only to supply energy
- Mitochondria to supply energy for synthesis of neurotransmitters and vesicles containing the chemical transmitter
- Receptors of the chemical transmitter
- Enzymes to hydrolyze the chemical transmitter

5. Temporal Summation:

- Results from stimulation of a single presynaptic fiber from different sites
- Always produces an inhibition of the postsynaptic cell
- Always produces an excitation of the postsynaptic cell
- Results from repeated stimulation of a single presynaptic fiber at a high rate

6. Acetylcholine is secreted by:

- All the postganglionic neurons of the sympathetic nervous system
- All preganglionic neurons of the autonomic nervous system
- The adrenal medulla

7. The site of contact between two neurons is called:

- Axons
- Synapse
- Cell body
- Dendrites

8. When drug molecules pass from the blood to the neuron which of the following cells will control their passage?

- Oligodendrocyte
- Microglia
- Astrocytes
- Schwann cells

9. Excitatory synapses (EPSP):

- Is due to the release of an excitatory transmitter such as ACh and catecholamines
- Is due to the release of GABA
- Decreases the permeability of postsynaptic membrane to Na^+ and to K^+

10. As regards synaptic transmission:

- It can occur in both directions
- Rapid repeated stimulation causes initial potentiation followed by fatigue
- Anesthetics inhibit synaptic transmission via closure of K^+ channels
- Hypoxia increases synaptic transmission and leads to convulsions

11. As regards serotonin, all of the following are true except:

- It is a small molecule that belongs to class II of synaptic transmitters
- It is an inhibitory transmitter for pain pathways in the spinal cord
- It is involved in sleep cycles and control of mood
- It is secreted by neurons of locus ceruleus in the pons

12. The frequency of discharge of action potentials is:

- Directly proportionate to the intensity of the stimulus
- Directly proportionate to logarithm of the intensity of the stimulus
- Inversely proportionate to the intensity of the stimulus
- Inversely proportionate to logarithm of the intensity of the stimulus

Answers :

1-d | 2-d | 3-c | 4-b | 5-d | 6-b | 7-b | 8-c | 9-a | 10-b | 11-d | 12-a

Dr. Selim

1. The synaptic knob contains:

- Mitochondria only to supply energy.
- Mitochondria to supply energy for synthesis of and vesicles containing the chemical transmitter.

- C. Receptors of the chemical transmitter.
- D. Enzymes to hydrolyze the chemical transmitter.

2. Excitatory synapses (EPSP):

- A. Is due to the release of an excitatory transmitter as ACh and catecholamines.
- B. Is due to the release of GABA.
- C. Decreases the permeability of postsynaptic membrane to Na^+ and to K^+ .
- D. Produces hyperpolarization of postsynaptic membrane.

3. Inhibitory synapses (IPSP):

- A. Is due to the release of an excitatory transmitter as ACh and catecholamines.
- B. Is due to the release of an inhibitory transmitter as GABA and Glycine.
- C. Decreases the permeability of postsynaptic membrane to K^+ and Cl^- .
- D. Produces polarization of postsynaptic membrane.

4. The presynaptic nerve impulse:

- A. Increases the permeability of the presynaptic terminal to Ca^{2+} .
- B. The release of transmitter into the synaptic cleft.
- C. Is needed to produce mEPSP.
- D. Reaches the terminal knobs as a wave of reversal polarity.

5. Temporal Summation:

- A. Results from stimulation of a single presynaptic fiber from different sites.
- B. Always produces an inhibition of the postsynaptic cell.
- C. Always produces an excitation of the postsynaptic cell.
- D. Results from repeated stimulation of a single presynaptic fiber at a high rate.

6. Spatial Summation:

- A. Results from repeated stimulation of a single presynaptic fiber at a high rate.
- B. Can't lead to excitation of the postsynaptic cell.
- C. Results from simultaneous stimulation of several excitatory presynaptic neurons that converge on the same postsynaptic neuron.
- D. Produces mIPSP.

7. Synaptic Delay:

- A. For a single synapse, it is about 0.8 ms.
- B. It is required for synthesis of the transmitter.
- C. The time required for conduction of a nerve impulse from the presynaptic to the postsynaptic neuron.

- D. It is decreased in polysynaptic reflex and increased in monosynaptic reflex.

8. Regarding the mechanism of synaptic transmission:

- A. Ca^{2+} ions are released from the presynaptic knob.
- B. Presynaptic knobs release only an excitatory transmitter.
- C. Chemical transmitter causes development of only excitatory postsynaptic potentials.
- D. Excitatory transmitter opens Na^+ channels.

9. Synaptic transmission is characterized by all of the following, EXCEPT:

- A. Impulses are transmitted from the pre- to the postsynaptic neurons.
- B. Alkalosis depresses synaptic transmission.
- C. Fatigue is due to exhaustion of neurotransmitter stores.
- D. Repetitive stimulation leads to synaptic potentiation.

10. Excitatory postsynaptic potential (EPSP):

- A. Is not propagated.
- B. Shows reversal of polarity.
- C. Lasts about 2 msec.
- D. May reach action potential at the soma.

11. An inhibitory neurotransmitter increases the permeability of the postsynaptic neuron to:

- A. K^+ .
- B. Na^+ .
- C. Ca^{2+} .
- D. HCO_3^- .

12. Regarding GABA, all of the following statements are true, EXCEPT:

- A. It hyperpolarizes the neurone producing an IPSP.
- B. It decreases Ca^{2+} influx into the neurone.
- C. It increases Cl^- influx and K^+ efflux.
- D. It is an excitatory neurotransmitter in the brain.

13. Synaptic transmission is facilitated by all the following, EXCEPT:

- A. Coffee and tea.
- B. Alkalosis.
- C. Strychnine.
- D. Hypoxia.

14. Drugs that facilitate Synaptic Transmission e.g. Caffeine:

- A. Increase transmitter destruction.
- B. Increase the release of excitatory transmitter.
- C. Increase the release of inhibitory transmitters.
- D. Na^+ influx and $\uparrow \text{K}^+$ outflux.

15. Acetylcholine is secreted by:

- A. All the postganglionic neurons of the sympathetic nervous system.
- B. All preganglionic neurons of the autonomic nervous system.
- C. The adrenal medulla.

Answers :

1-B | 2-A | 3-B | 4-D | 5-D | 6-C | 7-C | 8-D | 9-B | 10-A | 11-A | 12-D | 13-D | 14-B | 15-B

1. Connexons of gap junctions in electric synapses:

- a- are Ligand-gated
- b- are voltage-gated
- c- allow transmission of potential changes in both directions between the pre- and post-synaptic neurons
- d- close whenever the presynaptic neuron becomes hyperpolarized

2. Chemical synapses in the nervous system:

- a- allow diffusion of chemical substances from the presynaptic neuron into the postsynaptic neuron
- b- allow transmission of potential changes in one direction only; from the presynaptic to the postsynaptic neurons
- c- have potential-gated ionic channels
- d- are more numerous in the peripheral nervous system than the central nervous system

3. Synaptic cleft:

- a- is the space between two synapses on the surface of neurons
- b- allow diffusion of transmitters between neurons
- c- is filled with intracellular fluid
- d- ranges between 20 – 30 μm in width

4. Synaptic knobs:

- a- synthesize different types of neurotransmitters
- b- release neurotransmitters by diffusion across their membranes
- c- have ligand-gated Ca^{++} channels
- d- are located at the terminal end of dendrites

5. Synaptic transmission depends upon:

- a- direct transmission of impulses from the presynaptic neuron to the postsynaptic neuron
- b- diffusion of neurotransmitters from synaptic knobs into the soma and dendrites of postsynaptic neurons
- c- presence of voltage-gated Ca^{++} channels in membrane of synaptic knobs
- d- presence of voltage-gated Ca^{++} channels in the subsynaptic membrane

6. Postsynaptic receptors include all the following types, except:

- a- G-protein coupled receptors
- b- ligand-gated cation channels
- c- G-protein regulated K^{+} channels
- d- voltage-gated Cl^{-} channels

7. When opened, the ligand-gated cation channels do not allow diffusion of Cl^{-} because:

- a- the size of Cl^{-} is bigger than the bore of the channels
- b- intracellular negativity causes complete inhibition of Cl^{-} influx
- c- the channels are specific for diffusion of Na^{+} only
- d- the inner surface of the channels is negatively charged

8. Opening of ligand-gated Cl^{-} channels causes:

- a- inhibition of the postsynaptic neuron
- b- depolarization of the postsynaptic neuron
- c- initiation of an action potential
- d- block of ligand-gated cation channels

9. Small-molecule neurotransmitters include all the following types, except:

- a- Substance P
- b- gamma amino butyric acid
- c- acetylcholine
- d- norepinephrine

10. Neuropeptides:

- a- are synthesized in the synaptic knobs
- b- consist of small molecules
- c- interact with ligand-receptors
- d- could produce prolonged changes in the structure of postsynaptic neurons

Answers :

1-c | 2-b | 3-b | 4-a | 5-c | 6-d | 7-d | 8-a | 9-a | 10-d

Formatives & Mid

1. Acetylcholine is secreted specifically by all of the following EXCEPT:

- A. Terminals of large pyramidal cells from the motor cortex
- B. Preganglionic neurons of the autonomic nervous system
- C. Postganglionic neurons of the sympathetic nervous system
- D. Dopaminergic neurons in substantia nigra

2. Norepinephrine-secreting neurons located in the locus ceruleus:

- A. Control widespread areas of the brain to regulate wakefulness and mood
- B. Only act on the spinal cord
- C. Always inhibit target neurons
- D. Are part of parasympathetic system

3. The main effect of dopamine secreted from substantia nigra neurons is:

- A. Excitation
- B. Inhibition
- C. No effect
- D. Mixed excitation and inhibition

4. Glycine acts mainly as:

- A. Excitatory transmitter
- B. Inhibitory transmitter
- C. Mixed transmitter
- D. Modulator of mood

5. GABA is secreted in all of the following regions EXCEPT:

- A. Spinal cord
- B. Cerebellum
- C. Basal ganglia
- D. Locus ceruleus

6. Glutamate usually causes:

- A. Excitation
- B. Inhibition
- C. Modulation of calcium channels only
- D. Mixed inhibition and excitation

7. Serotonin secretion contributes mainly to:

- A. Excitation of motor neurons
- B. Inhibition of pain pathways and mood regulation
- C. Memory storage
- D. Sympathetic activation

8. Nitric oxide is different from other small-molecule transmitters because it:

- A. Is preformed in vesicles
- B. Is synthesized on demand and diffuses into postsynaptic neurons
- C. Only acts on ion channels
- D. Has short-term action only

9. Neuropeptides differ from small-molecule transmitters because:

- A. They are synthesized in presynaptic terminals
- B. They are synthesized in the neuronal cell body and released slowly
- C. They act faster than small-molecule transmitters
- D. They are recycled repeatedly

10. Divergence in a neuronal pool refers to:

- A. Multiple input neurons converging on one output neuron
- B. One input neuron synapsing with multiple output neurons
- C. Closed chain circuits
- D. After-discharge in interneurons

11. Convergence in a neuronal pool allows:

- A. Spreading of impulses from one neuron to many
- B. Spatial summation from multiple input neurons on a single output neuron
- C. After-discharge
- D. Reverberation

12. The Discharge Zone (DZ) in an Excitation Field (EF) is characterized by:

- A. Peripheral neurons receiving few terminals → partial depolarization
- B. Central neurons receiving many terminals → prolonged action potentials
- C. No response to input neuron
- D. Only inhibitory postsynaptic potentials

13. Synaptic delay is approximately:

- A. 0.5 ms
- B. 5 ms
- C. 50 ms
- D. 0.05 ms

14. Synaptic fatigue occurs due to:

- A. Repeated stimulation → depletion of presynaptic neurotransmitter
- B. Postsynaptic hyperpolarization
- C. Excessive release of neuropeptides
- D. Increased calcium influx

15. Synaptic potentiation (post-tetanic potentiation) results from:

- A. Increased presynaptic calcium → more neurotransmitter release
- B. Decreased presynaptic calcium
- C. Postsynaptic hyperpolarization
- D. Synaptic inhibition

Answers :

1-D | 2-A | 3-B | 4-B | 5-D | 6-A | 7-B | 8-B | 9-B | 10-B | 11-B | 12-B | 13-A | 14-A | 15-A |

Formative and Mid

1. The somatosensory system is responsible for:

- A- Motor control
- B- Sensory perception
- C- Hormonal regulation
- D- Memory consolidation

2. Sensory neurons carry information from:

- A- Brain to muscles
- B- Spinal cord to brain
- C- Receptor cells to CNS
- D- Endocrine glands to neurons

3. Which of the following is a type of somatosensory receptor?

- A- Thermoreceptor
- B- Photoreceptor
- C- Baroreceptor (for blood pressure)
- D- None of the above

4. Nociceptors respond to:

- A- Light
- B- Pain
- C- Temperature
- D- Pressure

5. Mechanoreceptors detect:

- A- Chemical changes
- B- Mechanical stimuli
- C- Temperature
- D- Pain

6. Chemoreceptors in the somatosensory system detect:

- A- Pressure
- B- Chemical stimuli
- C- Light
- D- Vibration

7. Thermoreceptors are specialized for:

- A- Heat and cold
- B- Pain
- C- Touch
- D- Sound

8. Sensory information from somatosensory receptors is first processed in:

- A- Muscle fibers
- B- Spinal cord
- C- Brainstem only
- D- Peripheral glands

9. Baroreceptors are responsible for detecting:

- A- Pressure changes
- B- Pain
- C- Chemical stimuli
- D- Temperature

10. Sensation can be:

- A- Only conscious
- B- Only unconscious
- C- Either conscious or unconscious
- D- None of the above

Answers:

1-B | 2-C | 3-A | 4-B | 5-B | 6-B | 7-A | 8-B | 9-A | 10-C

Dr. Selim

1. Free nerve endings are receptors of:

- A. Fine touch
- B. Vibration
- C. Pressure
- D. Pain

2. Which statement is correct:

- A. A β fibres are thickly myelinated and transmit nociceptive stimuli
- B. A δ fibres are unmyelinated and slower conducting than A β
- C. A δ fibres can alter the rate of firing depending on intensity and terminate in laminae I, IV & V
- D. C fibres are non-myelinated and rapid conducting

3. Fast pain is transmitted by:

- A. A β fibres
- B. A δ fibres
- C. C fibres
- D. Motor fibers

4. Delayed pain is transmitted by:

- A. A β fibres
- B. A δ fibres
- C. C fibres
- D. Motor fibers

5. Primary hyperalgesia:

- A. Occurs in healthy skin around the injured area
- B. Occurs in all site of skin showing redness
- C. Pain threshold not lowered
- D. Mechanism due to convergence-facilitation or lateral inhibition

6. Secondary hyperalgesia:

- A. Mechanism due to skin damage + PPS
- B. Pain threshold is lowered
- C. Occurs in healthy skin around the injured area
- D. Occurs in all site showing redness

7. About the sensory receptors:

- A. They are specialized structures at the peripheral ends of afferent neurons
- B. They detect changes in the surroundings
- C. They act as transducers
- D. All of the above

8. Stimulation of receptors leads to:

- A. Action potential
- B. Complete depolarization
- C. Non-specific response
- D. Partial depolarization

9. About the receptor potential, all the following are true EXCEPT:

- A. It has a longer duration than the action potential
- B. It is a state of partial depolarization due to increased Na^+ permeability
- C. It cannot be blocked by local anesthetic drugs
- D. It has a long refractory period

10. Receptor potential:

- A. Is a stage of partial depolarization
- B. Obey all-or-none law
- C. Is accompanied by absolute refractory period
- D. Cannot be summated

11. Receptor potential:

- A. Is a state of partial depolarization
- B. Is conducted actively to the adjacent node of Ranvier
- C. Can be graded but not summated
- D. Generates one action potential

12. Receptor (receptors):

- A. Adaptation means decreased amplitude of receptor potential & frequency of action potentials with constant stimulation
- B. Pain receptors are phasic receptors
- C. Touch receptors are tonic receptors
- D. Adaptation is caused by fatigue

13. A decrease in sensory response while a stimulus is maintained constant is due to:

- A. Adaptation
- B. Fatigue
- C. Graded response
- D. Compression

14. Adaptation in a sensory receptor is associated with:

- A. A decline in the amplitude of action potentials in the sensory nerve
- B. Reduction in intensity of the applied stimulus
- C. Decline in the amplitude of the generator potential
- D. Reduction in duration of sensory stimulus

15. Concerning adaptation of receptors:

- A. Decline in the rate of firing despite constant stimulation
- B. Occurs in all receptors to the same degree
- C. Accompanied by increased receptor potential
- D. Cold receptors adapt more rapidly than warmth receptors

16. An increase in the action potential frequency in a sensory nerve signifies:

- A. Increased intensity of the stimulus
- B. Cessation of stimulation
- C. Adaptation
- D. Increased conduction velocity

17. The mechanoreceptors perceive all the following sensations EXCEPT:

- A. Touch
- B. Pressure
- C. Vibration
- D. Pain

18. Sensation of warm is carried by:

- A. Type A alpha fibers
- B. Type A delta fibers
- C. Type B fibers
- D. Type C fibers

19. Lateral spinothalamic tract transmits:

- A. Fine touch
- B. Pain & temperature
- C. Vibration
- D. Kinesthetic sensation

20. The final destination of fine sensations:

- A. Gracile & cuneate tract
- B. Ventral nuclei of the thalamus
- C. Post-central gyrus of cerebral cortex
- D. Cerebellar cortex

21. The pain receptors:

- A. Are rapidly-adapting receptors
- B. Can be stimulated by overstimulation of any type of receptors
- C. Are connected to type C fibers only
- D. Are connected to thick myelinated A-delta fibers

22. Concerning pain, all are true EXCEPT:

- A. Threshold is the same for all persons
- B. Receptors are stimulated by bradykinin
- C. Pain is inhibited by stimulating A-fibers
- D. Paleospinothalamic tract inhibits reticular formation

Answers :

- 1-D | 2-C | 3-B | 4-C | 5-A | 6-C | 7-D | 8-D | 9-D |
10-A | 11-A | 12-A | 13-A | 14-C | 15-A | 16-A |
17-D | 18-D | 19-B | 20-C | 21-B | 22-A

1. Pain receptors in the skin are typically classified as which of the following?

- a. Encapsulated nerve ending
- b. Single class of morphologically specialized receptors
- c. Same type of receptor that detects position sense
- d. Free nerve endings

2. Free nerve endings are receptors of:

- a. Fine touch
- b. Vibration
- c. Pressure
- d. Pain

3. Deep pain is characterized by all the following EXCEPT:

- a. It is dull aching
- b. It is conducted by C non-myelinated fibers
- c. It is sharp localized prickling

4. Referred pain includes all the following EXCEPT:

- a. Cardiac pain
- b. Muscle pain
- c. Gastric pain
- d. Renal pain

5. Visceral pain is characterized by all the following EXCEPT:

- a. It is dull aching or colicky
- b. Transmitted by A δ fibers
- c. It is diffuse and not localized
- d. It is usually referred

6. Secondary hyperalgesia:

- a. Its mechanism may be due to skin damage and release of PPS
- b. The pain threshold is lowered
- c. Occurs in healthy skin around the injured area
- d. Occurs in all the site of skin area showing redness (Red flare)

7. Evidences supporting the gate theory for pain control are the following EXCEPT:

- a. Stress analgesia
- b. Reflex rigidity (immobilization)
- c. Rubbing of the skin around the injured area decreases pain sensation
- d. Morphine is used as a powerful analgesic drug

8. Which sensory receptor in the skin is NOT encapsulated?

- a. Ruffini endings
- b. Free nerve endings
- c. Pacinian corpuscle
- d. Meissner's corpuscles

9. Pacinian corpuscles are most responsive to:

- a. Light touch
- b. Pressure
- c. Pain
- d. Heat

10. Fast pain is transmitted by:

- a. A β fibres
- b. A δ fibres
- c. C fibres
- d. Motor fibres

11. Delayed pain is transmitted by:

- a. A β fibres
- b. A δ fibres
- c. C fibres
- d. Motor fibres

12. Primary hyperalgesia:

- a. Occurs in healthy skin around the injured area
- b. Occurs in all the site of skin area showing redness (Red flare)
- c. The pain threshold is not lowered
- d. Its mechanism may be due to convergence-facilitation or lateral inhibition

13. Secondary hyperalgesia:

- a. Its mechanism may be due to skin damage and release of PPS
- b. The pain threshold is lowered
- c. Occurs in healthy skin around the injured area
- d. Occurs in all the site of skin area showing redness (Red flare)

14. Pain threshold:

- a. Is the highest intensity of stimulus that causes pain
- b. Is the lowest intensity of stimulus that causes pain
- c. Differs markedly among different individuals
- d. Is elevated by substance P

15. The spinal pain gate is:

- a. Ligand controlled
- b. Voltage controlled
- c. Opened by enkephalin
- d. Closed by substance P

Answers :

1-D | 2-D | 3-C | 4-B | 5-B | 6-C | 7-B | 8-B | 9-B | 10-B | 11-C | 12-B | 13-C | 14-B | 15-C

1. Pain sensation:

- a) is evoked by strong stimulation of specific receptors
- b) produces reactions that block transmission of pain impulses
- c) arises from small encapsulated receptors
- d) occurs when the stimulus causes damage of the sensory receptors

2. Pain receptors:

- a) become more sensitive with prolonged stimulation
- b) are stimulated by prostaglandins
- c) are more numerous in viscera than other tissues
- d) include different morphological types

3. Pain threshold:

- a) is the highest intensity of stimulus that causes pain
- b) is the lowest intensity of stimulus that causes pain
- c) differs markedly among different individuals
- d) is elevated by substance P

4. Fast pain differs from slow pain in:

- a) being transmitted in the dorsal column pathway
- b) evoking a depressor autonomic reaction
- c) having a sharp quality
- d) arising from encapsulated pain receptors

5. Double pain sensation that is occasionally felt following painful stimulation of the skin is due to:

- a) repetition of the painful stimulus
- b) presence of dual pain pathways
- c) perception of pain at two different higher centers
- d) application of two painful stimuli simultaneously at two different sites

6. Cutaneous pain:

- a) is always sharp in character
- b) is transmitted by A δ sensory fibers
- c) is always followed by hyperalgesia
- d) evokes spasm of nearby muscles

7. Deep pain shows the following characteristics, except:

- a) dull aching
- b) throbbing
- c) evokes flexor reflexes
- d) diffuse

8. Visceral pain:

- a) is more common than the other types of pain
- b) arises only from wall of the visceral organs
- c) is often well localized
- d) evokes depressor autonomic reactions

9. Visceral pain is usually felt:

- a) deeply in the diseased viscera
- b) in deep tissues close to the diseased viscera
- c) in skin areas that just overlie the diseased viscera
- d) in skin areas remote from the diseased viscera

10. Pain produced by muscle spasm results from:

- a) mechanical stimulation of pain receptor by muscle spasm
- b) decreased release of lactic acid from the spastic muscle fibers
- c) release of compounds from the spastic muscle which increase the threshold for stimulation of pain receptors
- d) decreased oxygen supply to the muscle

11. Intracranial headache could result from painful stimuli applied on:

- a) the dura lining the inner surface of the bones of cranial vault
- b) the brain tissue
- c) wall of big intracranial veins
- d) arachnoid mater

12. Intracranial headache may result from all the following clinical causes, except:

- a) decreased CSF pressure
- b) spasm of scalp muscles
- c) distension of big venous sinuses
- d) pressure upon tentorium cerebelli

13. Reaction to pain includes all the following, except:

- a) increased heart rate
- b) depression
- c) withdrawal reflexes
- d) stoppage of impulse discharge from nociceptors in chronic painful conditions

14. The spinal pain gate is:

- a) Ligand controlled
- b) voltage controlled
- c) opened by enkephalin
- d) closed by substance

Answers:

1-d | 2-b | 3-b | 4-c | 5-b | 6-b | 7-c | 8-d | 9-d | 10-d
| 11-a | 12-b | 13-d | 14-a

1. Pain receptors in the skin are typically classified as which of the following?

- a. Encapsulated nerve ending
- b. Single class of morphologically specialized receptors
- c. Same type of receptor that detects position sense
- d. Free nerve endings

2. Pain receptors:

- a. Are rapidly adapting receptors
- b. Are free nerve endings
- c. Have low threshold
- d. Are not widely distributed

3. Free nerve endings are receptors of:

- a. Fine touch
- b. Vibration
- c. Pressure
- d. Pain

4. Free nerve endings are receptors of:

- a. Fine touch
- b. Vibration
- c. Pressure
- d. Pain

5. Which statement is correct:

- a. $A\beta$ fibres are thickly myelinated and normally transmit nociceptive stimuli
- b. $A\delta$ fibres are unmyelinated and slower conducting than $A\beta$ transmission
- c. $A\delta$ fibres can alter the rate of firing depending on the intensity of stimuli and terminate in laminae I, IV and V
- d. C fibres are non-myelinated

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- c. $A\delta$ fibers can alter the rate of firing depending on the intensity of stimuli and terminate in laminae I, IV, V
- d. C fibers are non-myelinated and rapid conducting

7. Pain threshold:

- a. Is the highest intensity of stimulus that causes pain
- b. Is the lowest intensity of stimulus that causes pain
- c. Differs markedly among different individuals
- d. Is elevated by substance P

8. Secondary hyperalgesia:

- a. Its mechanism may be due to skin damage and release of PPS

- b. The pain threshold is lowered
- c. Occurs in healthy skin around the injured area
- d. Occurs in all the site of skin area showing redness (red flare)

9. Deep pain is characterized by all the following EXCEPT:

- a. It is dull aching
- b. It is conducted by C non-myelinated fibers
- c. It is sharp localized prickling
- d. It is associated with body reactions (somatic, autonomic...)

10. Deep pain is characterized by all the following EXCEPT:

- a. It is dull aching
- b. It is conducted by C non-myelinated fibers
- c. It is sharp localized pricking
- d. It is associated with reflex immobilization

11. Visceral pain is characterized by all the following EXCEPT:

- a. It is dull aching or colicky
- b. Transmitted by $A\delta$ fibers
- c. It is diffuse and not localized
- d. It is usually referred

12. Visceral pain is characterized by all the following EXCEPT:

- a. It is dull aching or colicky
- b. Transmitted by $A\delta$ fibers
- c. It is diffuse and not localized
- d. It is usually referred

13. Referred pain includes all the following EXCEPT:

- a. Cardiac pain
- b. Muscle pain
- c. Gastric pain
- d. Renal pain

14. Referred pain includes all the following EXCEPT:

- a. Cardiac pain
- b. Muscle pain
- c. Gastric pain
- d. Renal pain

15. The spinal pain gate is:

- a. Ligand controlled
- b. Voltage controlled
- c. Opened by enkephalin
- d. Closed by substance P

16. Evidences supporting the gate theory for pain control are the following EXCEPT:

- a. Stress analgesia
- b. Reflex rigidity (immobilization)
- c. Rubbing of the skin around the injured area decreases pain sensation
- d. Morphine is used as a powerful analgesic drug

17. Evidences supporting the gate theory for pain control are the following EXCEPT:

- a. Stress analgesia
- b. Reflex rigidity (immobilization)
- c. Rubbing of the skin around the injured area decreases pain sensation
- d. Morphine is used as a powerful analgesic drug

18. Opiate receptors are present in all these sites EXCEPT:

- a. Periventricular neurons
- b. Periaqueductal neurons
- c. Substantia nigra
- d. AHCs

19. Intracranial headache could result from painful stimuli applied on:

- a. The dura lining the inner surface of the bones of cranial vault
- b. The brain tissue
- c. Wall of big intracranial veins
- d. Arachnoid mater

20. Regarding the causes of intracranial headache, select the FALSE one:

- a. Decreased CSF pressure
- b. Spasm of scalp muscles
- c. Distension of big venous sinuses
- d. Pressure upon tentorium cerebelli

Answers :

1-d | 2-b | 3-d | 4-d | 5-d | 6-c | 7-b | 8-c | 9-c | 10-c | 11-b | 12-b | 13-b | 14-b | 15-d | 16-b | 17-b | 18-d | 19-c | 20-b

1. Evidences supporting the gate theory for pain control are the following EXCEPT:

- a. Stress analgesia
- b. Reflex rigidity (immobilization)
- c. Rubbing of the skin around the injured area decreases pain
- d. Morphine is used as a powerful analgesic drug

2. The spinal pain gate is:

- a. Ligand controlled
- b. Voltage controlled
- c. Opened by enkephalin
- d. Closed by substance P

3. Enkephalin blocks pain transmission by:

- a. Blocking response of pain receptors to painful stimuli
- b. Slowing down transmission of pain impulses through synapses in the pain pathway
- c. Inhibiting the response of cerebral cortical somatic sensory area to pain signals
- d. Blocking Ca^{++} channels in the central terminals of pain sensory fibers

4. Opiate receptors are present in all these sites EXCEPT:

- a. Periventricular neurons
- b. Periaqueductal neurons
- c. Substantia nigra
- d. Anterior horn cells (AHCs)

5. Deep pain is characterized by all the following EXCEPT:

- a. Dull aching
- b. Conducted by C nonmyelinated fibers
- c. Sharp localized prickling
- d. Associated with body reactions (somatic, autonomic)

6. Intermittent claudication evokes:

- a. Visceral pain
- b. Deep pain
- c. Cutaneous hyperalgesia
- d. Colicky pain

7. Visceral pain is characterized by all the following EXCEPT:

- a. Dull aching or colicky
- b. Transmitted by $A\delta$ fibers
- c. Diffuse and not localized
- d. Usually referred

8. Stimuli which evoke true visceral pain include all the following EXCEPT:

- a. Sharp cutting
- b. Stretching
- c. Spasm
- d. Chemical irritation

9. Visceral pain is usually felt:

- a. Deeply in the diseased viscera
- b. In deep tissues close to the diseased viscera
- c. In skin areas that just overlie the diseased viscera
- d. In skin areas remote from the diseased viscera

10. Referred pain includes all the following EXCEPT:

- a. Cardiac pain
- b. Muscle pain
- c. Gastric pain
- d. Renal pain

11. The basis of referred pain is:

- a. Visceral pain signals from the skin synapse with separate neurons in dorsal horn
- b. Visceral and skin pain transmission is received by a common set of neurons in the thalamus
- c. Visceral pain signals are rarely sufficient to activate dorsal horn neurons
- d. Some visceral and skin pain signals converge on a common set of neurons in the dorsal horn

12. Pain receptors in the skin are typically classified as:

- a. Encapsulated nerve endings
- b. Single class of morphologically specialized receptors
- c. Same type of receptor that detects position sense
- d. Free nerve endings

13. Pain threshold:

- a. Highest intensity of stimulus that causes pain
- b. Lowest intensity of stimulus that causes pain
- c. Differs markedly among different individuals
- d. Is elevated by substance P

14. Substance most likely associated with production of pain after injury:

- a. Acetylcholine
- b. Adrenaline
- c. GABA
- d. Substance P

15. Fast pain differs from slow pain in:

- a. Being transmitted in dorsal column pathway
- b. Evoking depressor autonomic reaction
- c. Having a sharp quality
- d. Arising from encapsulated pain receptors

16. Double pain sensation following skin stimulation is due to:

- a. Repetition of the painful stimulus
- b. Presence of dual pain pathways
- c. Perception of pain at two different higher centers
- d. application of two painful stimuli simultaneously at two different sites

17. Primary cutaneous hyperalgesia:

- a. develops in the normal skin region around the area of flare
- b. is an abnormal condition in the skin in which painful stimuli become more severe
- c. is due to changes in threshold of pain receptors
- d. is associated with throbbing type of pain

18. Secondary Hyperalgesia:

- a. Its mechanism may be due to skin damage and release of PPS.
- b. The pain threshold is lowered.
- c. Occurs in healthy skin around the injured area
- d. Occurs in all the site of skin area showing redness (Red Flare)

19. Intracranial headache could result from painful stimuli applied on:

- a. the dura lining the inner surface of the bones of cranial vault
- b. the brain tissue
- c. wall of big intracranial veins
- d. arachnoid mater

20. Intracranial headache may result from all the following clinical causes, except:

- a. decreased CSF pressure
- b. spasm of scalp muscles
- c. distension of big venous sinuses
- d. pressure upon tentorium cerebelli

Answers :

- 1.b | 2. d | 3. d | 4. d | 5. c | 6. b | 7. b | 8. a | 9. d |
10. b | 11. d | 12. d | 13. b | 14. d | 15. c | 16. B |
17. b | 18. c | 19. c | 20. b

Formatives & Dr.Selim

1. The gamma (γ) efferent motor fibers:

- a. Supply the central part of the intrafusal muscle fibers.
- b. Supply the extrafusal muscle fibers.
- c. Are axons of the large AHCs (α AHCs).
- d. Supply the peripheral contractile part of the intrafusal muscle fibers.

2. Maximal stimulation and discharge of the muscle spindle occurs when:

- a. Muscle contraction occurs without $\uparrow \gamma$ efferent discharge.
- b. The muscle is stretched and there is $\uparrow \gamma$ efferent discharge.
- c. There is $\uparrow \gamma$ efferent discharge.
- d. Tapping on the muscle tendon with a hammer.

3. What is the primary mechanism of action of tricyclic antidepressants (TCAs)?

- a. Increasing dopamine levels
- b. Blocking serotonin and norepinephrine reuptake
- c. Inhibiting monoamine oxidase enzymes
- d. Enhancing GABAergic activity

4. What is the common side effect of SSRI antidepressants?

- a. Hypertension
- b. Sexual dysfunction
- c. Increased appetite
- d. Hypoglycemia

5. The gamma (γ) efferent motor fibers:

- a. Supply the central part of the intrafusal muscle fibers.
- b. Supply the extrafusal muscle fibers.
- c. Are axons of the large AHCs (α AHCs).
- d. Supply the peripheral contractile part of the intrafusal muscle fibers.

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- a. Muscle contraction occurs without $\uparrow \gamma$ efferent discharge.
- b. The muscle is stretched and there is $\uparrow \gamma$ efferent discharge.
- c. There is $\uparrow \gamma$ efferent discharge.
- d. Tapping on the muscle tendon with a hammer.

7. The dynamic response of the stretch reflex is characterized by the following EXCEPT:

- a. Its afferents are the primary endings.
- b. The response is maintained subtetanic contraction.
- c. The stimulus is sudden stretch.
- d. The receptor is nuclear bag fibers.

8. The supraspinal facilitatory centers controlling stretch reflex are the following EXCEPT:

- a. The vestibular nuclei.
- b. The facilitatory reticular formation.
- c. The neocerebellum.
- d. The basal ganglia.

9. Higher motor commands originate in all the following centers, EXCEPT:

- a. Cerebral cortex
- b. Thalamus
- c. Caudate nucleus
- d. Cerebellum

10. A reflex arc includes:

- a. At least two sets of sequential neurons
- b. At least two sequential sets of central synapses
- c. At least two types of sensory receptors
- d. At least two types of efferent neurons

11. Recruitment of a reflex response is due to:

- a. Difference in the amount of presynaptic inputs to the various efferent neurons initiating the reflex
- b. Difference in the conduction velocity of the various afferent neurons mediating the reflex
- c. Delay at the neuromuscular junction
- d. Presence of inhibitory interneurons in the reflex pathway

12. After-discharge of reflex responses:

- a. Increase the magnitude of the reflex responses
- b. Delays the onset of fatigue of reflex responses
- c. Involves interneuron circuits
- d. Depends upon spatial summation

13. Central delay of reflex actions:

- a. Is due to presence of inhibitory interneurons in the reflex pathway
- b. Is the time between stimulation of receptors and response of the effector muscle
- c. Is longer in the flexor reflex than in the stretch reflex
- d. Is determined by the velocity of conduction of impulses along the afferent and efferent neurons

14. Fatigue of reflexes:

- a. Develops gradually and recovers rapidly
- b. Develops rapidly and recovers slowly
- c. Develops gradually and recovers slowly
- d. Develops rapidly and recovers rapidly

15. Muscle spindles:

- a. Are found in all skeletal muscles
- b. Are found only in large skeletal muscles
- c. Consist of small numbers of extrafusal muscle fibers
- d. Consist of a large number of extrafusal muscle fibers

16. The nuclear-bag fibers of muscle spindles are innervated by:

- a. $A\gamma$ nerve fibers
- b. $A\beta$ nerve fibers
- c. $A\delta$ nerve fibers
- d. Ia nerve fibers

17. The nuclear-chain fibers of spindles are innervated by:

- a. $A\alpha$ and $A\delta$ nerve fibers
- b. $A\delta$ and C nerve fibers
- c. Ia and II nerve fibers
- d. Only type II nerve fibers

18. The central ends of afferents from muscle spindles synapse with all the following types of neurons, EXCEPT:

- a. α -motor neurons of the same muscle
- b. γ -motor neurons of the same muscle
- c. Local interneurons

19. γ -motor neuron discharge to a muscle is inhibited by impulses reaching the γ -motor neurons from all the following sources, EXCEPT:

- a. Golgi tendon organs of the same muscle
- b. Spindles of antagonistic muscles
- c. Medullary reticular formation
- d. Pontine reticular formation

20. When the γ -motor neuron discharge to a muscle decreases, it causes:

- a. Increased muscle spindle discharge
- b. Shortening of the spindle fibers
- c. Stronger contraction of the extrafusal muscle fibers
- d. Decreased spindle sensitivity to stretch

Answers :

1. D | 2. B | 3. B | 4. B | 5. D | 6. B | 7. B | 8. D | 9. B |
 10. A | 11. A | 12. C | 13. C | 14. C | 15. A | 16. D |
 17. C | 18. B | 19. D | 20. D

Formatives

1. The tendon jerk is lost (areflexia) if:

- a. The reflex arc is injured at any facilitatory point.
- b. The supraspinal facilitatory centers or their descending tracts are affected.
- c. The supraspinal inhibitory center or their descending tracts are affected.
- d. Increase muscle tone.

2. Clonus occurs when:

- a. Tapping on the tendon of the muscle in normal person.
- b. Tapping on the tendon of the muscle in hypotonia.
- c. Tapping on the tendon of the muscle in hypertonia.
- d. There is maintained stretch on the muscle tendon in cases of hypertonia.

3. The center of the knee tendon jerk :

- a. C4-C5
- b. S1-S2
- c. L3-L4
- d. C5-C6

Previous Exams

4. Tendon jerks are clinically examined to assess

- a- integrity of muscle spindles
- b- integrity of reflex pathway
- c- the total reflex time of the jerk
- d- central delay time of the jerk

5. Exaggeration of tendon jerks could result from any of the following conditions, except :-

- a- lesions of supraspinal facilitatory centers
- b- lesions of supraspinal inhibitory centers
- c- increased γ -motor neuron discharge
- d- anxiety

6. Clonus :-

- a- is a sign of decreased supraspinal facilitation
- b- initiated by briefly stretching the tendon of the muscle
- c- is manifested as oscillating mechanical vibrations following tendon jerks
- d- associates exaggeration of tendon jerks

7. Stretch reflex is characterized by the following except :-

- a- disynaptic reflex
- b- high localization
- c- shows reciprocal innervations.
- d- it is of graded response

8. Stretch of an innervated muscle evokes :-

- a- contraction of its spindles
- b- contraction of its extrafusal fibers
- c- contraction of antagonistic muscles
- d- relaxation of synergistic muscles

9. When a skeletal muscle is suddenly stretched :-

- a- it relaxes suddenly
- b- it develops a static stretch reflex
- c- it develops a dynamic stretch reflex
- d- it develops clonic contractions

10. The shortest reflex time is recorded with :-

- a- a flexor withdrawal reflex
- b- an inverse stretch reflex
- c- a stretch reflex
- d- a scratch reflex

11. Inverse stretch reflex :-

- a- increases the possibility of avulsion of the excessively stretched muscle from its bony attachments
- b- has no reciprocal innervation circuits
- c- is clinically manifested by lengthening reaction
- d- is clinically tested by examining the tendon jerks

12. A tendon jerk :-

- a- is a dynamic stretch reflex
- b- is a static stretch reflex
- c- is evoked by gradually stretching the muscle
- d- is evoked by stimulation of tendon receptors

13. The tendon jerk which has its center in the 5th and 6th cervical segments of the spinal cord is :-

- a- the jaw jerk
- b- the deltoid jerk
- c- the biceps jerk
- d- the triceps jerk

14. Absence of a tendon jerk could result from any of the following conditions, except :-

- a- lesions of supraspinal facilitatory centers
- b- lesions of the efferent neurons
- c- lesions of the afferent neurons
- d- lesions of the spinal nerve centers

15. Exaggeration of tendon jerks could result from any of the following conditions, except :-

- a- lesions of supraspinal facilitatory centers
- b- lesions of supraspinal inhibitory centers
- c- increased γ -motor neuron discharge
- d- anxiety

16. The inverse stretch reflex ?

A- Occurs when 1a spindle afferents are inhibited

B- Is a monosynaptic reflex initiated by activation of the Golgi tendon organ

C- Uses type II afferent fibers from the Golgi tendon organ

D- Is a disynaptic reflex with an excitatory interneuron inserted between the afferent and efferent limbs

17. Withdrawal reflexes are ?

A- Initiated by nocuous stimulation of the skin

B- Can lead to the appearance of clonus

C- An example of a stretch reflex

D- Accompanied by the same response on both sides of the body

ANSWERS:

1- a | 2- D | 3-C | 4- B | 5-A | 6-D | 7-A | 8-B | 9-C |
10-C | 11-C | 12-A | 13-C | 14-A | 15-A | 16-C |
17-A

Formative

1. Which of the following is the precursor for Serotonin (5-HT) synthesis?
 - a) Tyrosine
 - b) Histidine
 - c) Tryptophan
 - d) Phenylalanine
2. The major excitatory neurotransmitter in the CNS is:
 - a) Glycine
 - b) GABA
 - c) Glutamate
 - d) Taurine
3. After three weeks of starvation the energy of the brain is derived from oxidation of :
 - a) proteins
 - b) ketone bodies
 - c) glucose
 - d) glycogen
4. Energy derived from carbohydrate oxidation is :
 - a) 4 Kcal
 - b) 9 Kcal
 - c) 10 Kcal
 - d) 12 Kcal

Mid

5. Ammonia toxicity in the brain is primarily due to the depletion of:
 - a) Glutamine
 - b) GABA
 - c) Alpha-ketoglutarate
 - d) Glutamate

Quizzes

6. Glutamate is converted to the inhibitory neurotransmitter GABA by:
 - a) Glutamate decarboxylase
 - b) Glutamate dehydrogenase
 - c) Glutaminase
 - d) GABA transaminase
7. The rate-limiting enzyme in the synthesis of catecholamines (Dopamine, NE) is:
 - a) DOPA decarboxylase
 - b) Tyrosine hydroxylase
 - c) Phenylethanolamine N-methyltransferase
 - d) MAO
8. Parkinson's disease is characterized by the loss of neurons that produce:
 - a) Acetylcholine
 - b) Dopamine
 - c) Serotonin
 - d) Histamine

9. Which enzyme requires Pyridoxal phosphate (Vitamin B6) as a cofactor for GABA synthesis?
 - a) Glutaminase
 - b) Glutamate decarboxylase
 - c) Glutamine synthetase
 - d) LDH
10. Which of the following neurotransmitters is synthesized from Histidine?
 - a) Melatonin
 - b) Histamine
 - c) Epinephrine
 - d) Dopamine
11. In the brain, ammonia is detoxified primarily by forming:
 - a) Urea
 - b) Uric acid
 - c) Glutamine
 - d) Creatinine

ANSWERS :

1-c | 2-c | 3-b | 4-a | 5-c | 6-a | 7-b | 8-b | 9-b | 10-b | 11-

Formatives

1. Regarding the antidepressant imipramine:

- a) It acts by inhibiting reuptake of serotonin and norepinephrine
- b) It is selective serotonin reuptake inhibitor
- c) It inhibits monoamine oxidase enzyme
- d) Dose adjustment is only required in presence of renal diseases

2. Which of the following antidepressants may be used in smoking cessation:

- a) Mirtazapine
- b) Phenelzine
- c) Duloxetine
- d) Bupropion

3. Most of antidepressants produce their effects:

- a) Soon after administration
- b) A day later
- c) After 3-4 weeks
- d) After 6 months

4. Priapism is a possible side effect of:

- a) Amitriptyline
- b) Trazodone
- c) Mirtazapine
- d) Bupropion

5. Which of the following is most likely to cause orthostatic hypotension:

- a) Desipramine
- b) Fluoxetine
- c) Venlafaxine
- d) Sertraline

6. Administration of MAOI would most likely cause changes in the CNS concentration of:

- a) Ach
- b) Histamine
- c) Norepinephrine
- d) GABA

7. Most of antidepressants produce their effects:

- a) Soon after administration
- b) A day later
- c) After 3-4 weeks
- d) After 6 months

ANSWERS

1-a | 2-d | 3-c | 4-b | 5-a | 6-c | 7-c

1. The following is a selective serotonin reuptake inhibitor used in treatment of psych depression:

- a- Imipramine
- b- Amitriptyline
- c- Fluoxetine
- d- Maprotiline
- e- Tranylcypromine

2. The following statements are correct about MAO inhibitors EXCEPT:

- a- Their effects persist for more than a week after cessation of administration
- b- Combination of old cheese or beer with MAO inhibitors can cause hypertensive crisis
- c- Selegiline (deprenyl) is a selective MAO (B) inhibitor
- d- Combination of a MAO-inhibitor and a selective serotonin reuptake inhibitor produces a synergistic effect useful in treatment of depression
- e- MAO-inhibitors reduce oxidative deamination of noradrenaline, serotonin & dopamine

3. All of the following statements about fluoxetine are correct EXCEPT:

- a- It is a selective serotonin reuptake inhibitor
- b- It is used in treatment of epilepsy
- c- It may cause anxiety and insomnia
- d- It has anticholinergic and cardiotoxic effects
- e- It may cause sexual dysfunction

4. Psychic depression can be treated by the following:

- a- Desipramine
- b- Tranylcypromine
- c- Amineptine
- d- Fluoxetine
- e- All of the above

5. Side effects of MAO inhibitors include the following EXCEPT:

- a- Cheese reaction
- b- Hepatic microsomal enzyme induction
- c- Hepatotoxicity
- d- Delayed onset of action
- e- None of the above

6. The following are useful antidepressants EXCEPT:

- a- Amitriptyline
- b- Fluoxetine
- c- Isoniazid
- d- Imipramine
- e- Tranylcypromine

7. Tricyclic antidepressants produce the following EXCEPT:

- a- Block dopamine receptors
- b- Block serotonin receptors
- c- Block muscarinic receptors
- d- Block histamine receptors
- e- None of the above

8. Tricyclic antidepressants are useful in the following EXCEPT:

- a- Depression and phobias
- b- Cardiac arrhythmias
- c- Migraine prophylaxis
- d- Nocturnal enuresis
- e- Obsessive neurosis

9. Side effects of tricyclic antidepressants include the following EXCEPT:

- a- Sedation
- b- Cardiac arrhythmias
- c- Anti-cholinergic effects
- d- Hypertension
- e- Cholestatic jaundice

10. Side effects of MAO inhibitors include:

- a- Hepatotoxicity
- b- Severe hypertension with tyramine
- c- Insomnia and tremors
- d- All of the above
- e- None of the above

11. Selective serotonin reuptake inhibitors produce:

- a- Anti-depressant action
- b- Anti-anxiety action
- c- Potent atropine-like action
- d- Appetite stimulation
- e- Antihypertensive action

12. Paroxetine is:

- a- Dopamine agonist
- b- Antihypertensive drug
- c- Has atropine-like action
- d- Selective serotonin reuptake inhibitor
- e- Antispasmodic drug

13. Both the older (tricyclic) antidepressants and the newer ones (eg, fluoxetine) relieve signs and symptoms of depression mainly by doing which one of the following things?

- a- Blocking adrenergic and serotonin receptors in the extrapyramidal system
- b- Blocking muscarinic receptors in the brain
- c- Blocking reuptake of one or more neurotransmitters in the brain
- d- Selectively stimulating serotonin receptors
- e- Stimulating the synthesis of GABA

14. Selective serotonin reuptake inhibitors is:

- a- Fluoxetine
- b- Paroxetine
- c- Sertraline
- d- Bupropion
- e- Citalopram

15. A 56-year-old male is on numerous medications, many of which are metabolized by the cytochrome P-450 system. He went to the psychiatric with depression. Given his multiple medications, which of the following antidepressant drugs should be avoided?

- a- Fluoxetine
- b- Citalopram
- c- Sertraline
- d- Fluvoxamine
- e- Imipramine

16. A 56-year-old male has been smoking 2 packs of cigarettes a day for the last 20 years. He has tried nicotine patches with no success and wants to know if there is any pill he could try. What medication could the physician recommend?

- a- Citalopram
- b- Phenelzine
- c- Buspirone
- d- Bupropion
- e- Imipramine

17. A 10-year-old boy has nocturnal enuresis. The physician writes an order for a low dose of imipramine. After a couple of weeks on the drug, the episodes of bed-wetting decrease dramatically. Which of the following best accounts for the beneficial effects of the drug in this patient?

- a- Alleviate depression signs and symptoms by increasing neuronal catecholamine reuptake
- b- Blocks muscarinic receptors in the bladder
- c- Causes sedation such that the boy sleeps without voiding
- d- Reduces renal blood flow, glomerular filtration and urine output
- e- Releases antidiuretic hormone

Answers :

1-c | 2-d | 3-b | 4-e | 5-b | 6-c | 7-a | 8-b | 9-d | 10-d | 11-a | 12-d | 13-c | 14-d | 15-a | 16-d | 17-b

Formatives

1. Example of local anesthetic with intrinsic sympathomimetic vasoconstrictive properties

- a) Mepivacaine
- b) Cocaine
- c) Procaine
- d) Lidocaine

2. Inclusion of a vasoconstrictor in the local anesthetic solution is recommended to:

- a) Terminate its action
- b) Increase the systemic absorption
- c) Bloodless surgical field
- d) Decrease the duration of action

3. As regards local anesthetics, which of the following statements is correct:

- a) Local anesthetics are acidic compounds its activity are strongly pH-dependent
- b) Elimination of amide groups by plasma esterases
- c) Inflamed tissues are somewhat sensitive to local anesthetic agents
- d) Bupivacaine may produce serious arrhythmia and tachycardia

4. One of the anesthetic drugs is commonly used topically & parenterally and may be used as antiarrhythmic:

- a) Bupivacaine
- b) Lidocaine
- c) Mepivacaine
- d) Cocaine

5. Amal has a history of allergy to ester type of local anesthetics, an acceptable choice for producing nerve block anesthesia will be:

- a) Chloroprocaine
- b) Mepivacaine
- c) Procaine
- d) Tetracaine

Answers :

1-b | 2-c | 3-d | 4-b | 5-b

1. Cocaine produces all the following actions EXCEPT:

- A. Local anaesthetic
- B. CNS stimulation
- C. Quinidine-like
- D. Miosis
- E. CNS stimulant

2. Side effects of local anesthetics include:

- A. Allergy especially with ester type
- B. CNS stimulation followed by depression
- C. Persistent anesthetic effect

D. All of the above

E. None of the above

3. Which of the following local anesthetics is an amide?

- A. Cocaine
- B. Procaine
- C. Dibucaine
- D. Tetracaine
- E. Benzocaine

4. Which of the following is a local anesthetic of very low solubility useful only for surface anaesthesia?

- A. Procaine
- B. Benzocaine
- C. Lidocaine
- D. Tetracaine
- E. Prilocaine

5. All local anesthetics produce vasodilatation EXCEPT:

- A. Procaine
- B. Tetracaine
- C. Cocaine
- D. Lidocaine
- E. None of the above

6. The following is a local anesthetic amide that has in addition an anti-arrhythmic effect:

- A. Cocaine
- B. Tetracaine
- C. Lidocaine
- D. Benzocaine
- E. Mepivacaine

7. Local anesthetic agents include which of the following:

- A. Lidocaine
- B. Procaine
- C. Bupivacaine
- D. Cocaine
- E. All of the above

8. Local anesthetic action is related to which of the following?

- A. Blockade of voltage-dependent sodium channels
- B. Blockade of voltage-dependent calcium channels
- C. Frequency of nerve cell firing
- D. Resting membrane potential of the nerves
- E. All of the above

Answers :

1-D | 2-D | 3-C | 4-B | 5-C | 6-C | 7-E | 8-E

1. Local anesthetics block impulse conduction along nerve axons by targeting which type of channels?

- a) K⁺ channels
- b) Ca²⁺ channels
- c) Na⁺ channels
- d) Cl⁻ channels

2. Which of the following is NOT a characteristic of an ideal local anesthetic?

- a) Rapid onset
- b) Long duration of action
- c) Selective blockade of motor nerves
- d) Minimal local tissue irritation and no systemic toxicities

3. How does pH affect the penetration of local anesthetic agents?

- a) Penetration is increased at alkaline pH
- b) Penetration is increased at acidic pH
- c) Penetration is not affected by pH
- d) Penetration is poor at both alkaline and acidic pH

4. What is the main site of action for local anesthetics?

- a) Axon terminal
- b) Cell nucleus
- c) Cell membrane
- d) Mitochondria

5. The action of local anesthetic can be prolonged by:

- a) Administration by continuous intravenous infusion
- b) Using phenylephrine
- c) Giving 3 or more local anesthetics together
- d) Administration by inhalation
- e) Using atropine

6. Which organ is primarily responsible for the metabolism of ester-type local anesthetics?

- a) Liver
- b) Kidneys
- c) Lungs
- d) Brain

7. Which type of nerve fibers are more susceptible to the action of local anesthetics?

- a) Myelinated fibers
- b) Small non-myelinated fibers
- c) Motor axons
- d) Large diameter fibers

8. The potency of local anesthetics is positively correlated with:

- a) Molecular weight
- b) Lipid solubility
- c) Protein binding affinity
- d) Water solubility

9. What is the primary reason for adding a vasoconstrictor to local anesthetics in clinical practice?

- a) To increase systemic toxicity
- b) To decrease the rate of absorption
- c) To enhance nerve conduction
- d) To induce vasodilation

10. Which of the following is NOT a service performed by vasoconstrictors added to local anesthetics?

- a) Localization of the anesthetic at the desired site
- b) Reduction of systemic toxicity
- c) Promotion of wound healing
- d) Decrease in bleeding during an operation

Answers :

1-c | 2-c | 3-a | 4-c | 5-b | 6-a | 7-b | 8-b | 9-b | 10-c

Formatives

1. **Naegleria fowleri** infection is acquired by:
 - A. Swimming in fresh water.
 - B. Drinking of contaminated water.
 - C. Walking bare footed on moist sand soil.
 - D. Blood transfusion.
2. Which parasitic stage is responsible for amoebic brain abscess?
 - A. Trophozoite.
 - B. Uninucleate cyst.
 - C. Binucleate cyst.
 - D. Quadrinucleate cyst.
3. The diagnostic stage of granulomatous amoebic encephalitis (G.A.E) is:
 - a. Trophozoite stage.
 - b. Cyst stage.
 - c. Trophozoite and cyst.
4. The infective stage in case of Negleriasis is:
 - a. Trophozoite.
 - b. Cyst.
 - c. Trophozoite and cyst.

Previous Exams

5. The gold standard for laboratory confirmation of malaria is:
 - a. Blood smear
 - b. Urine smear
 - c. C.S.F smear
 - d. Sputum smear
6. Which of the following is the most common cause of sleeping disorders? (Note: In the context of this parasite):
 - a. Taenia solium
 - b. Trypanosoma brucei
 - c. Naegleria fowleri
 - d. Plasmodium falciparum
7. Plasmodium falciparum malaria stage or stages that can be seen in blood film:
 - a. Trophozoites
 - b. Schizonts and gametocytes
 - c. Schizonts
 - d. Ring and gametocytes
8. Avoidance of swimming or bathing in contaminated water is essential for prevention and control of:
 - a. African sleeping sickness
 - b. Brain amoebic abscess
 - c. Malignant falciparum malaria
 - d. Primary amoebic meningoencephalitis
9. Amphotericin B is used in the treatment of:
 - a. Leishmania spp.
 - b. Entamoeba histolytica
 - c. Trypanosoma brucei
 - d. Naegleria fowleri

10. The portal of entry for Naegleria fowleri to reach the brain is:
 - A. Through the mouth by ingestion
 - B. Through the skin by penetration
 - C. Through the olfactory neuroepithelium in the nose
 - D. Through blood transfusion
11. Primary Amoebic Meningoencephalitis (P.A.M) is caused by:
 - A. Acanthamoeba species
 - B. Naegleria fowleri
 - C. Entamoeba histolytica
 - D. Toxoplasma gondii
12. Which of the following is the most common habitat for Naegleria fowleri?
 - A. Salt water (oceans)
 - B. Warm fresh water (lakes and swimming pools)
 - C. Dry desert sand
 - D. Treated tap water only
13. In patients with G.A.E (Granulomatous Amoebic Encephalitis), infection usually occurs in:
 - A. Previously healthy children
 - B. Athletes and swimmers
 - C. Immunocompromised or debilitated patients
 - D. Neonates only
14. Acanthamoeba can reach the Central Nervous System (CNS) through:
 - A. Direct spread from the ear
 - B. Hematogenous spread from primary skin or lung lesions
 - C. Direct penetration of the skull
 - D. Ingestion of cysts in food
15. The clinical course of Primary Amoebic Meningoencephalitis (P.A.M) is typically:
 - A. Chronic and lasts for years
 - B. Subacute with gradual onset
 - C. Rapidly progressive and fatal within days
 - D. Always asymptomatic
16. Which protozoa is known for causing "Keratitis" in contact lens wearers, which may precede CNS involvement?
 - A. Naegleria
 - B. Acanthamoeba
 - C. Giardia
 - D. Trichomonas

17. For the diagnosis of Naegleria fowleri, the specimen of choice for microscopic examination is:

- A. Stool sample
- B. Blood film
- C. Cerebrospinal fluid (CSF)
- D. Urine sample

18. Which of the following stages of Naegleria fowleri has "Pseudopodia" and is found in the CSF?

- A. Cyst stage
- B. Flagellated stage
- C. Trophozoite stage
- D. Egg stage

19. Granulomatous Amoebic Encephalitis (G.A.E) is characterized by:

- A. Acute suppurative inflammation
- B. Formation of granulomas containing both trophozoites and cysts
- C. Absence of any inflammatory response
- D. Liquefactive necrosis without any cellular infiltration

ANSWERS:

1-A | 2-A | 3-C | 4-A | 5-a | 6-b | 7-d | 8-d | 9-d |
10-C | 11-B | 12-B | 13-C | 14-B | 15-C | 16-B |
17-C | 18-C | 19-B

Formatives

1. Cerebral abscess

- a- spread mainly by lymphatic spread
- b- Caused by aerobic organism
- c- Can initiate seizures
- d- It has a specific microscopic picture

2. Bacterial meningitis cause

- a- Decrease in level of CSF glucose
- b- Decrease in cell count
- c- Increase of cell count specially lymphocytes
- d- Clear in appearance

3. Infection of the brain parenchyma is called:

- a- Meningitis
- b- Encephalitis
- c- Brain edema
- d- Astrocytoma

4. Bacteria can reach the brain through

- a- Direct spread
- b- Lymphatic spread
- c- Transperitoneal spread
- d- Trans pleural spread

5. Aseptic meningitis can be caused by:

- A. Gram positive bacteria
- B. Virus
- C. Immunoglobulins
- D. Gram negative bacteria

6. The predominant cell in CSF in viral meningitis is:

- a) Neutrophils
- b) Macrophages
- c) Plasma cells
- d) Lymphocytes

7. The most common cause of meningitis in neonates is:

- a) Hemophilus influenza
- b) Neisseria meningitidis
- c) E. coli
- d) Streptococcus pneumoniae

8. Viral meningitis

- a- It is a fatal condition
- b- caused mainly by enteroviruses
- c- Lead to severe neurological dysfunction
- d- There is of monocytes in the CSF

9. Bacterial meningitis

- a- Indicates prompt initiation of antibiotic treatment.
- b- Resolve without specific treatment
- c- Is not a medical emergency
- d- Indicates prompt initiation of antiviral treatment

10. Most cases of encephalitis are mostly caused by

- a- Meningococci

- b- Hepatitis C virus

- c- Herpes simplex virus

- d- Mumps virus

11. Diagnosis of meningitis is made through the analysis of CSF. Which of the following results would be expected with a diagnosis of bacterial meningitis?

- a- Cloudy CSF, elevated WBC, elevated protein and decreased glucose

- b- Clear CSF, decreased WBC, elevated protein and elevated glucose

- c- Cloudy CSF, decreased WBC, decreased protein and elevated glucose

- d- Clear CSF, elevated WBC, decreased protein and decreased glucose

Answers :

1. c | 2. a | 3. b | 4. a | 5. b | 6. d | 7. c | 8. b | 9. a | 10. c | 11. A

1. The commonest cause of neonatal meningitis is E. coli

- A. True

- B. False

2. Acute lymphocytic meningitis is usually of viral infection

- A. True

- B. False

3. Neglected suppurative otitis media may predispose to meningitis

- A. True

- B. False

4. Brain abscess is always pyaemic

- A. True

- B. False

5. CSF in acute septic meningitis does NOT show

- A. Increased amount

- B. Increased protein content

- C. Increased sugar content

- D. Increased polymorphonuclear cells

- E. Organisms

6. CSF shows cob web coagulum of fibrin on standing in case of

- A. Acute pyogenic meningitis

- B. Chronic meningitis

- C. Brain abscess

- D. Encephalitis

- E. Cerebral infarction

Answers :

- 1-A | 2-A | 3-A | 4-B | 5-C | 6-B